To Convert from	*To*	*Multiply by*	*Notes*
Volume [meter^3, m^3]			
λ ($1\lambda = 1$ μL $= 1 \times 10^{-6}$ L)	m^3	1×10^{-9}	E
Foot^3, ft^3	m^3	$2.831\,684\,659\,2 \times 10^{-2}$	E
Gallon (U.S. liquid), gal	m^3	$3.785\,411\,784 \times 10^{-3}$	E
Inch^3, in^3	m^3	$1.638\,706\,4 \times 10^{-5}$	E
Liter, L	m^3	1×10^{-3}	E
Ounce (U.S. fluid), oz	m^3	$2.957\,352\,956\,25 \times 10^{-5}$	E
Pint (U.S. liquid), pt	m^3	$4.731\,764\,73 \times 10^{-4}$	E
Quart, qt	m^3	$9.463\,529\,46 \times 10^{-4}$	E
Stere (st)	m^3	1	E
Yard^3, yd^3	m^3	0.764 554 857 984	E
Work (see Energy)			

Notes for the preceding table:

AA Values from *Astronomical Almanac* [2011].

AQ Values from *Allen's Astrophysical Quantities* [Cox, 2000].

C Values from CODATA [2006 website].

D Values from exact quantities, rounded off to 13 significant figures.

E (Exact) indicates that the conversion is exact by definition of the non-SI unit or that it is obtained from other exact conversions.

IAU Values from the International Astronomical Union [2009 website].

IERS From the IERS [McCarthy and Petit, 2004].

NIST Values from the National Institute of Standards and Technology [Thompson and Taylor, 2008].

(1) Care should be taken in transforming magnetic units, because the dimensionality of magnetic quantities (**B**, **H**, etc.) depends on the system of units. Most of the conversions given here are between SI and EMU (electromagnetic). The following equations hold in both sets of units:

$\mathbf{T} = \mathbf{m} \times \mathbf{B} = \mathbf{d} \times \mathbf{H}$

$\mathbf{B} = \mu\,\mathbf{H}$

$\mathbf{m} = I\mathbf{A}$ for a current loop in a plane

$\mathbf{d} = \mu\,\mathbf{m}$

with the following definitions:

$\mathbf{T} \equiv$ torque

$\mathbf{B} \equiv$ magnetic induction (commonly called "magnetic field")

$\mathbf{H} \equiv$ magnetic field strength or magnetic intensity

$\mathbf{m} \equiv$ magnetic moment

$I \equiv$ current loop

$\mathbf{A} \equiv$ vector normal to the plane of the current loop (in the direction of the angular velocity vector of the current loop) with magnitude equal to the area of the loop.

$\mathbf{d} \equiv$ magnetic dipole moment

$\mu \equiv$ magnetic permeability

The permeability of vacuum, μ_0, has the following values, by definition:

$\mu_0 \equiv 1$ (dimensionless), EMU

$\mu_0 \equiv 4\pi \times 10^{-7}$ N/A^2, SI

Therefore, in electromagnetic units in vacuum, magnetic induction and magnetic field strength are equivalent and the magnetic moment and magnetic dipole moment are equivalent. For practical magnetostatics, space is a vacuum but the spacecraft itself may have $\mu \neq \mu_0$.

Value of Useful Mathematical Constants

π	≈3.141 592 653 589 793 238 462 643
e	≈2.718 281 828 459 045 235 360 287
e^π	≈23.140 692 632 779 269 006
$\log_{10}x$	≈0.434 294 481 903 251 827 651 128 9 $\log_e x$
$\log_e x$	≈2.302 585 092 994 045 684 017 991 $\log_{10} x$
$\log_e \pi$	≈1.144 729 885 849 400 174 143 427

Values are from *The Handbook of Mathematical Functions, with Formulas, Graphs, and Mathematical Tables* [Abramowitz and Stegun, 1970]

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Appendix I
Earth Satellite Parameters

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Explanation

Table I-1 provides a variety of quantitative data for Earth-orbiting satellites. Limitations, formulas, and text references are given below. The independent parameter in the formulas is the distance, r, from the center of the Earth in km. The outer column on each table is the altitude, $h \equiv r - R_E$, where $R_E = 6{,}378.1366$ km is the equatorial radius of the Earth.

1. *Instantaneous Access Area for a 0 deg Elevation Angle* or the *Full Geometric Horizon* (10^6 km^2). All the area that an instrument or antenna could potentially see at any instant if it were scanned through its normal range of orientations for which the spacecraft elevation is above 0 deg [Eq. (C-29)].
2. *Instantaneous Access Area for a 5 deg Minimum Elevation Angle* (10^6 km^2) = same as col.1 but with elevation of 5 deg.
3. *Instantaneous Access Area for a 10 deg Minimum Elevation Angle* (10^6 km^2) = same as col.1 but with elevation of 10 deg.
4. *Instantaneous Access Area for a 20 deg Minimum Elevation Angle* (10^6 km^2) = same as col.1 but with elevation of 20 deg.
5. *Area Access Rate for an Elevation of 0 deg* (10^3 km^2/s) = the rate at which new land is coming into the spacecraft's access area [Eq. (C-30)].
6. *Area Access Rate for an Elevation Limit of 5 deg* (10^3 km^2/s) = same as col. 5 with an elevation of 5 deg.
7. *Area Access Rate for an Elevation Limit of 10 deg* (10^3 km^2/s) = same as col. 5 with an elevation of 10 deg.
8. *Area Access Rate for an Elevation Limit of 20 deg* (10^3 km^2/s) = same as col. 5 with an elevation of 20 deg.
9. *Maximum Time in View for a Satellite Visible to a Minimum Elevation Angle of 0 deg* (min) = $P\lambda_{max}$/180 deg, where P is from col. 69 and λ_{max} is from col. 13. Assume a circular orbit over a nonrotating Earth [Eq. (C-31)].
10. *Maximum Time in View for a Satellite Visible to a Minimum Elevation Angle of 5 deg* (min) = same as col. 9 with λ_{max} for a 5 deg taken from col. 14.
11. *Maximum Time in View for a Satellite Visible to a Minimum Elevation Angle of 10 deg* (min) = same as col. 9 with λ_{max} for a 10 deg taken from col. 15.
12. *Maximum Time in View for a Satellite Visible to a Minimum Elevation Angle of 20 deg* (min) = same as col. 9 with λ_{max} for a 20 deg taken from col. 16.
13. *Earth Central Angle for a Satellite at 0 deg Elevation* (deg) = *Maximum Earth Central Angle* = acos(R_E/r). Alternatively, Maximum Earth Central Angle = 90−ρ, where ρ is from col. 25 [Eqs. (8-25) and (C-28)].
14. *Earth Central Angle for a Satellite at 5 deg Elevation* (deg) = 90−ε−η, where η = asin (cosε sinρ), ρ is from col. 66 and ε = 5 deg [Eqs. (8-27), (8-28), and (8-29)].
15. *Earth Central Angle for a Satellite at 10 deg Elevation* (deg) = same as col. 14 but with ε = 10 deg.
16. *Earth Central Angle for a Satellite at 20 deg Elevation* (deg) = same as col. 14 but with ε = 20 deg.
17. *Range to Horizon* = *Range to a satellite at 0 deg elevation* (km) = $(r^2 - R_E{}^2)^{1/2}$, where R_E = 6,378.14 km is the equatorial radius of the Earth [Eq. (C-27) and Fig. 8-28].
18. *Range to a Satellite at 5 deg Elevation* (km) = *Maximum Range for Satellites with a Minimum Elevation Angle of 5 deg* (km) = R_E (sinλ / sinη), where R_E = 6,378.1366 km is the equatorial radius of the Earth, λ is from col. 14, η = 90 deg − λ − ε, and ε = 5 deg [Eq. (8-30)].
19. *Range to a Satellite at 10 deg Elevation* (km) = same as col. 18 with λ from col. 15 and ε = 10 deg.
20. *Range to a Satellite at 20 deg Elevation* (km) = same as col. 18 with λ from col. 16 and ε = 20 deg.
21. *Nadir Angle for a Satellite at 0 deg Elevation Angle* (deg) = *Max. Nadir Angle for Any Point on the Earth* = *Earth Angular Radius* = asin (R_E / r), where R_E = 6,378.1366 km is the equatorial radius of the Earth [Eq. (8-25)].
22. *Nadir Angle for a Satellite at 5 deg Elevation Angle* (deg) = *Maximum Nadir Angle for Points on the Ground with a Minimum Elevation Angle of 5 deg* = 90 deg − ε − λ, where ε = 5 deg and λ is the Earth central angle from col. 14 [Eq. (8-29)].
23. *Nadir Angle for a Satellite at 10 deg Elevation Angle* (deg) = same as col. 22 with ε = 10 deg.
24. *Nadir Angle for a Satellite at 20 deg Elevation Angle* (deg) = same as col. 22 with ε = 20 deg.
25. *Euler Rotation Rate* (deg/min) = Angular rate of rotation about the Euler axis = ω_{orbit} sin i / sin *EACL*, where ω_{orbit} is the orbital angular velocity from col. 63, *EACL* is the Euler Axis Co-Latitude from

col. 32, and i is the assumed inclination of 0 deg for this column [Eq. (C-25)].

26–31. *Euler Rotation Rate* (deg/min) = same as col. 25 with value of the inclination at the top of the column and Euler Axis Co-Latitude from the corresponding cols. 32–38.

32. *Euler Axis Co-Latitude* (deg) = atan $[\omega_{orbit} \sin i / (-\omega_{earth} + \omega_{orbit} \cos i)]$, where ω_{orbit} is the orbital angular velocity from col. 63, ω_{earth} is the Earth's angular velocity on its axis = 0.250684446 deg/min, and i is the assumed inclintion of 0 deg for this column [Eqs. (8-79) and (C-24)].

33–38. *Euler Axis Co-Latitude* (deg) = same as col. 32 with value of the inclination at the top of the column.

39. *Inertial Rotation Rate* (deg/min) = Orbit angular rate = same as col. 63. Data is repeated here for convenient comparison with cols. 26–31.

40. *Minimum Atmospheric Scale Height* (km) = RT / Mg, where R is the molar gas constant, T is the temperature, M is the mean molecular weight, and g is the standard gravitational acceleration (see inside front cover). The value is based on the MSIS atmospheric model [Hedin, 1987, 1988, and 1991]. The solar flux value, F10.7, was chosen such that 10% of all measured data are less than this value (65.8×10^{-22} W·m^{-2}·Hz^{-1}). See Sec. 9.4.4.

41. *Mean Atmospheric Scale Height* (km) = same as col. 40 but with a mean F10.7 value of 118.7×10^{-22} W·m^{-2}·Hz^{-1}.

42. *Maximum Atmospheric Scale Height* (km) = same as col. 40 but with an F10.7 value of 189.0×10^{-22} W·m^{-2}·Hz^{-1}. This is the F10.7 value such that 10% of all measured values are above it.

43. *Minimum Atmospheric Density* (kg/m^3), from MSIS atmospheric model [Hedin, 1987, 1988, and 1991]. The solar flux value, F10.7, was chosen such that 10% of all measured data are less than this value (65.8×10^{-22} W·m^{-2}·Hz^{-1}). See Sec. 9.4.4. The MSIS model is limited to the region between 90 and 2,000 km. Below 150 km and above 600 km the error increases because less data have been used. All data have been averaged across the Earth with a 30 deg step size in longitude and 20 deg steps in latitude (–80 deg, to +80 deg). This over-represents the Earth's polar regions; however, satellites spend a larger fraction of their time at high latitudes. The solar hour angle was adapted to the individual location on the Earth with UT = 12.00 Noon.

44. *Mean Atmospheric Density* (kg/m^3) = same as col. 43 but with a mean F10.7 value of 118.7×10^{-22} W·m^{-2}·Hz^{-1}.

45. *Maximum Atmospheric Density* (kg/m^3) = same as col. 43 but with an F10.7 value of 189.0×10^{-22} W·m^{-2}·Hz^{-1}. This is the F10.7 value such that 10% of all measured values are above it.

46. *Minimum ΔV to Maintain Altitude at Solar Minimum* (m/s per year) = $\pi (C_D A/m) \times \rho v/P$, where ρ is from col. 43, v is from col. 62, P is from col. 69 expressed in years, and the ballistic coefficient, $m/C_D A$, is assumed to be 50 kg/m^2. ΔV estimates are not meaningful above 1,500 km [Eq. (9-36)].

47. *Maximum ΔV to Maintain Altitude at Solar Maximum* (m/s per year) = same as col. 46 with ρ from col. 45; Ballistic coefficient $m/C_D A = 50$ kg/m^2.

48. *Minimum ΔV to Maintain Altitude at Solar Minimum* (m/s per year) = same as col. 46 with ρ from col. 43; Ballistic coefficient $m/C_D A = 200$ kg/m^2.

49. *Maximum ΔV to Maintain Altitude at Solar Maximum* (m/s per year) = same as col. 46 with ρ from col. 45; Ballistic coefficient $m/C_D A = 200$ kg/m^2.

50. *Orbit Decay Rate at Solar Minimum* (km/year) = $-2\pi (C_D A/m) \rho r^2/P$, where ρ is from col. 43, P is from col. 69 (expressed in years), and the ballistic coefficient, $m/C_D A$, is assumed to be 50 kg/m^2. Orbit decay rates are not meaningful above 1,500 km [Eq. (9-34)].

51. *Orbit Decay Rate at Solar Maximum* (km/year) = same as col. 50, with ρ from col. 45 and the ballistic coefficient, $m/C_D A$, assumed to be 50 kg/m^2.

52. *Orbit Decay Rate at Solar Minimum* (km/year) = same as col. 50 with ρ from col. 43, and the ballistic coefficient, $m/C_D A$, assumed to be 200 kg/m^2.

53. *Orbit Decay Rate at Solar Maximum* (km/year) = same as col. 50, with ρ from col. 45 and the ballistic coefficient, $m/C_D A$, assumed to be 200 kg/m^2.

54. *Estimated Orbit Lifetime at Solar Minimum* (days) = Data was produced using the software package SatLife. Ballistic coefficient, $m/C_D A$, assumed to be 50 kg/m^2.

55. *Estimated Orbit Lifetime at Solar Maximum* (days) = same as col. 54 with the ballistic coefficient, $m/C_D A$, assumed to be 50 kg/m^2.

56. *Estimated Orbit Lifetime at Solar Minimum* (days) = same as col. 54 with the ballistic coefficient, $m/C_D A$, assumed to be 200 kg/m^2.

57. *Estimated Orbit Lifetime at Solar Maximum* (days) = same as col. 54 with the ballistic coefficient, $m/C_D A$, assumed to be 200 kg/m^2.

58. *Plane Change ΔV* [(km/s)/deg] = 2 v_{circ} sin (0.5 deg), where v_{circ} is from col. 62. Assumes circular orbit and linear sine function; [Eq. (9-78)]

59. *ΔV Required for a 1 km Altitude Change* (m/s) = assumes a Hohmann Transfer with $r_B - r_A = 1$ km; [Eqs. (9-69), and (9-70)].

60. *Maximum Angular Rate As Seen from a Ground Station* (deg/s) = $2\pi r/hP$, where $h \equiv r - R_E$ is the

altitude and P is from col. 69. This is the angular rate as seen from the surface of a non-rotating Earth of a satellite in a circular orbit passing directly overhead. (See col. 63 for the angular velocity as seen from the center of the Earth.) [Eqs. (8-74) and (C-32)].

61. *Sun-Synchronous Inclination* (deg) = acos ($-4.773\ 513\ 28 \times 10^{-15}\ r^{7/2}$); assumes circular orbit with node rotation rate of 0.985609 deg/day to follow the mean motion of the Sun. Above 6,000 km altitude there are no Sun synchronous circular orbits [Eq. (C-34)].

62. *Circular Velocity* (km/s) = $(\mu_E/r)^{1/2}$ = $631.348\ 110\ r^{-1/2}$ [Eq. (9-8)].

63. *Orbit Angular Velocity* (deg/min) = $360/P = 2.170\ 414\ 925 \times 10^6 r^{-3/2}$, where P is from col. 69. This is the angular velocity with respect to the center of the Earth for a circular orbit. (See col. 60 for angular rate with respect to ground stations) [Eqs. (9-15) and (9-17)].

64. *Escape Velocity* (km/s) = $(2\mu/r)^{1/2}$ = $892.861\ 059\ r^{-1/2} = (2)^{1/2} \times v_{circ}$ [Eq. (9-7)].

65. *ΔV Required to De-Orbit* (m/s) = the velocity change needed to transform the assumed circular orbit to an elliptical orbit with an unchanged apogee and a perigee of 50 km [Eqs. (9-85a), and (9-85b)]

66. *Angular Radius of the Earth* (deg) = asin (R_E / r), where R_E = 6,378.14 km is the equatorial radius of the Earth [Eq. (8-25)].

67. *One Degree Field of View Mapped onto the Earth's Surface at Nadir from Altitude h* (km) = The length on the Earth's curved surface of a 1 deg arc projected at nadir from this altitude. Note: This data is *very* nonlinear [Eqs. (8-27), (8-28), and (8-29)].

68. *Range to Horizon* (km) = same as col. 17 = $(r^2 - R_E{}^2)^{1/2}$, where R_E = 6,378.1366 km is the equatorial radius of the Earth. For the range to points other than the true horizon (i.e., $\varepsilon \neq 0$ deg) use cols. 18, 19, and 20 [Eq. (8-30)].

69. *Period* (min) = $1.658\ 669\ 0213 \times 10^{-4}\ r^{3/2} = (1/60) \times 2\pi\ (r^3/\mu)^{1/2}$. Assumes a circular orbit, r is measured in km, and μ = 398,600.4356 km^3/s^2. Note that period is the same for an eccentric orbit with semimajor axis = r; [Eq. (9-4b)].

70. *Maximum Eclipse* (minutes) = $(\rho/180\ \text{deg})P$, where ρ is from col. 66 and P is from col. 69. This is the maximum eclipse for a circular orbit. Eclipses at this altitude in an eccentric orbit can be longer. [Eq. (C-37)]

71. *Revolutions per Day* (#) = $1{,}436.068\ 34/P$, where P is from col. 69. Note that this is revolutions per sidereal day, where the *sidereal day* is the day relative to the fixed stars which is approximately 4 minutes shorter than the solar day of 1,440 minutes. [Eq. (C-35)]

72. *Node Spacing* (deg) = 360 deg × (P / 1,436.068 34), where P is from col. 69. This is the spacing in longitude between successive ascending or descending nodes for a satellite in a circular orbit [Eq. (C-36)].

73. *Node Precession Rate* (deg/day) = $-2.064\ 745\ 722 \times 10^{14}\ r^{-7/2} \cos i = -1.5\ n\ J_2\ (R_E/a)^2\ (\cos i)\ (1 - e^2)^{-2}$, where i is the inclination, e the eccentricity (which is set to zero), n is the mean motion [$= (\mu/a^3)^{1/2}$], a the semimajor axis, and J_2 the dominant zonal coefficient in the expansion of the Legendre polynomial describing the geopotential. This is the angle through which the orbit rotates in inertial space in a 24 hour period. Assumes a circular orbit; r is in km in the first expression [Eq. (C-33)].



For other altitudes, plots of the data, or orbits around other celestrial bodies, see the book website.

References



For annotated bibliography of Appendix I, see the book website.

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Table I-1. Earth Satellite Parameters.

	1	2	3	4	5	6	7	8	9	10	11	12
	INSTANTANEOUS ACCESS AREA				AREA ACCESS RATE				MAXIMUM TIME IN VIEW			
Alt. (km)	0 deg elevation (10^6 km^2)	5 deg elevation (10^6 km^2)	10 deg elevation (10^6 km^2)	20 deg elevation (10^6 km^2)	0 deg elevation (10^3 km^2/s)	5 deg elevation (10^3 km^2/s)	10 deg elevation (10^3 km^2/s)	20 deg elevation (10^3 km^2/s)	0 deg elevation (min)	5 deg elevation (min)	10 deg elevation (min)	20 deg elevation (min)
0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100	3.95	1.51	0.67	0.21	17.24	10.71	7.15	3.96	4.84	3.00	2.00	1.11
150	5.87	2.66	1.31	0.44	20.76	14.01	9.86	5.69	5.98	4.02	2.82	1.63
175	6.83	3.26	1.67	0.58	22.23	15.42	11.07	6.50	6.49	4.48	3.21	1.88
200	7.77	3.89	2.06	0.73	23.56	16.73	12.20	7.28	6.96	4.92	3.58	2.13
225	8.71	4.52	2.47	0.90	24.78	17.93	13.26	8.03	7.42	5.34	3.94	2.39
250	9.64	5.17	2.89	1.08	25.90	19.04	14.27	8.76	7.85	5.74	4.29	2.62
275	10.57	5.82	3.32	1.28	26.93	20.08	15.21	9.45	8.27	6.13	4.62	2.87
300	11.48	6.48	3.77	1.48	27.89	21.06	16.11	10.12	8.67	6.50	4.95	3.10
325	12.39	7.14	4.23	1.70	28.79	21.97	16.96	10.76	9.06	6.87	5.28	3.34
350	13.30	7.81	4.70	1.92	29.63	22.83	17.76	11.38	9.44	7.22	5.59	3.57
375	14.19	8.48	5.17	2.15	30.42	23.65	18.53	11.98	9.81	7.57	5.90	3.80
400	15.08	9.15	5.66	2.39	31.15	24.41	19.26	12.56	10.17	7.91	6.21	4.03
450	16.85	10.50	6.64	2.89	32.50	25.83	20.62	13.66	10.88	8.57	6.80	4.49
500	18.58	11.85	7.64	3.42	33.71	27.10	21.86	14.68	11.55	9.21	7.38	4.93
550	20.29	13.20	8.66	3.97	34.78	28.25	22.99	15.63	12.21	9.83	7.95	5.37
600	21.98	14.54	9.69	4.53	35.75	29.30	24.03	16.52	12.86	10.43	8.50	5.81
650	23.64	15.88	10.72	5.12	36.62	30.25	24.99	17.36	13.48	11.02	9.04	6.24
700	25.28	17.21	11.76	5.72	37.40	31.12	25.87	18.14	14.10	11.60	9.58	6.66
750	26.89	18.53	12.81	6.33	38.11	31.91	26.68	18.87	14.71	12.17	10.10	7.08
800	28.49	19.85	13.85	6.95	38.75	32.63	27.43	19.56	15.30	12.74	10.62	7.50
850	30.06	21.15	14.90	7.58	39.33	33.29	28.12	20.20	15.89	13.29	11.13	7.92
900	31.61	22.44	15.94	8.21	39.85	33.90	28.76	20.80	16.48	13.84	11.64	8.33
950	33.14	23.73	16.98	8.86	40.32	34.45	29.35	21.37	17.05	14.38	12.14	8.74
1,000	34.64	25.00	18.02	9.51	40.74	34.96	29.89	21.90	17.62	14.92	12.64	9.15
1,100	37.60	27.51	20.09	10.82	41.47	35.85	30.86	22.86	18.75	15.99	13.63	9.97
1,200	40.47	29.97	22.13	12.14	42.05	36.58	31.69	23.71	19.87	17.04	14.60	10.78
1,300	43.28	32.38	24.15	13.47	42.51	37.19	32.40	24.46	20.97	18.07	15.57	11.58
1,400	46.01	34.75	26.15	14.80	42.86	37.70	32.99	25.11	22.07	19.11	16.53	12.39
1,500	48.67	37.07	28.12	16.14	43.12	38.10	33.50	25.69	23.16	20.13	17.48	13.19
2,000	61.02	47.97	37.51	22.69	43.43	39.07	34.93	27.59	28.56	25.21	22.21	17.19
2,500	71.98	57.81	46.14	28.96	42.71	38.89	35.19	28.39	33.98	30.30	26.96	21.23
3,000	81.77	66.70	54.05	34.86	41.47	38.10	34.78	28.52	39.46	35.44	31.76	25.34
3,500	90.57	74.76	61.29	40.38	39.95	36.98	33.99	28.22	45.04	40.68	36.65	29.53
4,000	98.52	82.10	67.92	45.52	38.33	35.68	32.98	27.68	50.74	46.03	41.65	33.83
4,500	105.74	88.79	74.02	50.31	36.68	34.30	31.86	26.97	56.56	51.50	46.77	38.23
5,000	112.32	94.93	79.63	54.77	35.05	32.92	30.70	26.18	62.52	57.10	52.00	42.74
6,000	123.90	105.78	89.62	62.81	31.97	30.22	28.37	24.49	74.85	68.67	62.82	52.08
7,000	133.74	115.06	98.23	69.84	29.18	27.73	26.16	22.80	87.73	80.77	74.14	61.86
8,000	142.22	123.10	105.71	76.02	26.70	25.48	24.13	21.20	101.15	93.37	85.94	72.07
9,000	149.59	130.11	112.27	81.48	24.51	23.47	22.30	19.72	115.10	106.48	98.22	82.70
10,000	156.06	136.29	118.07	86.35	22.57	21.67	20.66	18.36	129.56	120.08	110.96	93.74
15,000	179.34	158.65	139.22	104.35	15.69	15.21	14.63	13.25	209.23	195.03	181.21	154.73
20,000	193.80	172.65	152.56	115.89	11.63	11.34	10.97	10.04	300.06	280.54	261.44	224.50
20,184*	194.23	173.06	152.95	116.24	11.52	11.23	10.87	9.95	303.60	283.87	264.57	227.22
25,000	203.65	182.23	161.73	123.91	9.05	8.85	8.60	7.92	400.90	375.52	350.60	302.13
30,000	210.79	189.19	168.42	129.80	7.29	7.15	6.96	6.44	510.87	479.15	447.92	386.92
35,786†	216.94	195.21	174.21	134.93	5.86	5.76	5.62	5.23	648.62	609.00	569.90	493.27
378,022‡	251.36	229.12	207.11	164.45	0.22	0.21	0.21	0.20	19,556	18,459	17,363	15,176

*GPS Altitude †GEO ‡Moon and L4, L5 Lagrange Points

See front of table for formulas and sources. See the book website for an Excel version of this table that allows you to plot the variables vs. altitude, to adjust the altitude range under consideration, and to determine these parameters for orbits around other celestrial bodies.

Table I-1. Earth Satellite Parameters. Continued

13	14	15	16	17	18	19	20	21	22	23	24	
EARTH CENTRAL ANGLE				RANGE				NADIR ANGLE				
0 deg elevation (deg)	5 deg elevation (deg)	10 deg elevation (deg)	20 deg elevation (deg)	0 deg elevation (km)	5 deg elevation (km)	10 deg elevation (km)	20 deg elevation (km)	0 deg elevation (deg)	5 deg elevation (deg)	10 deg elevation (deg)	20 deg elevation (deg)	Alt. (km)
0.00	0.00	0.00	0.00	0	0	0	0	90.00	85.00	80.00	70.00	0
10.06	6.24	4.16	2.30	1,134	707	477	277	79.92	78.76	75.84	67.70	100
12.31	8.27	5.81	3.35	1,391	942	671	406	77.69	76.73	74.19	66.65	150
13.27	9.17	6.56	3.85	1,504	1,048	761	468	76.73	75.83	73.44	66.15	175
14.16	10.00	7.28	4.34	1,610	1,147	846	530	75.84	75.00	72.72	65.66	200
15.00	10.79	7.96	4.81	1,709	1,241	929	590	75.00	74.21	72.04	65.19	225
15.79	11.54	8.62	5.28	1,803	1,331	1,009	649	74.21	73.46	71.38	64.72	250
16.53	12.25	9.25	5.73	1,893	1,417	1,086	707	73.47	72.75	70.75	64.27	275
17.24	12.93	9.85	6.17	1,979	1,500	1,160	764	72.76	72.07	70.15	63.83	300
17.91	13.58	10.44	6.60	2,062	1,580	1,233	820	72.09	71.42	69.56	63.40	325
18.56	14.20	11.00	7.02	2,142	1,657	1,304	876	71.44	70.80	69.00	62.98	350
19.18	14.80	11.55	7.44	2,219	1,732	1,373	930	70.82	70.20	68.45	62.56	375
19.78	15.38	12.08	7.84	2,294	1,805	1,440	984	70.22	69.62	67.92	62.16	400
20.92	16.48	13.09	8.63	2,438	1,944	1,570	1,090	69.08	68.52	66.91	61.37	450
21.98	17.52	14.05	9.38	2,575	2,078	1,695	1,193	68.02	67.48	65.95	60.62	500
22.98	18.49	14.96	10.11	2,705	2,206	1,816	1,294	67.02	66.51	65.04	59.89	550
23.93	19.42	15.82	10.81	2,831	2,329	1,932	1,392	66.07	65.58	64.18	59.19	600
24.84	20.30	16.65	11.48	2,952	2,448	2,045	1,489	65.16	64.70	63.35	58.52	650
25.70	21.15	17.45	12.14	3,069	2,563	2,155	1,584	64.30	63.85	62.55	57.86	700
26.52	21.95	18.21	12.77	3,183	2,675	2,262	1,677	63.48	63.05	61.79	57.23	750
27.31	22.73	18.95	13.39	3,293	2,784	2,367	1,769	62.69	62.27	61.05	56.61	800
28.07	23.47	19.66	13.98	3,401	2,890	2,469	1,859	61.93	61.53	60.34	56.02	850
28.80	24.19	20.34	14.56	3,506	2,994	2,569	1,948	61.20	60.81	59.66	55.44	900
29.50	24.88	21.00	15.13	3,608	3,095	2,667	2,035	60.50	60.12	59.00	54.87	950
30.18	25.55	21.64	15.68	3,709	3,194	2,763	2,121	59.82	59.45	58.36	54.32	1,000
31.47	26.83	22.87	16.73	3,904	3,388	2,951	2,291	58.53	58.17	57.13	53.27	1,100
32.69	28.02	24.02	17.73	4,092	3,574	3,132	2,456	57.31	56.98	55.98	52.27	1,200
33.83	29.15	25.11	18.69	4,275	3,755	3,308	2,618	56.17	55.85	54.89	51.31	1,300
34.91	30.23	26.14	19.60	4,452	3,931	3,480	2,776	55.09	54.77	53.86	50.40	1,400
35.94	31.24	27.13	20.47	4,624	4,102	3,648	2,932	54.06	53.76	52.87	49.53	1,500
40.42	35.68	31.43	24.33	5,433	4,905	4,437	3,673	49.58	49.32	48.57	45.67	2,000
44.08	39.30	34.97	27.54	6,176	5,645	5,167	4,368	45.92	45.70	45.03	42.46	2,500
47.15	42.35	37.95	30.28	6,875	6,342	5,856	5,032	42.85	42.65	42.05	39.72	3,000
49.78	44.97	40.52	32.65	7,543	7,008	6,516	5,671	40.22	40.03	39.48	37.35	3,500
52.08	47.25	42.75	34.72	8,187	7,650	7,154	6,291	37.92	37.75	37.25	35.28	4,000
54.10	49.26	44.73	36.57	8,812	8,274	7,774	6,897	35.90	35.74	35.27	33.43	4,500
55.91	51.05	46.49	38.21	9,422	8,883	8,380	7,490	34.09	33.95	33.51	31.79	5,000
58.98	54.12	49.51	41.04	10,608	10,067	9,558	8,649	31.02	30.88	30.49	28.96	6,000
61.53	56.64	52.00	43.38	11,760	11,217	10,704	9,779	28.47	28.36	28.00	26.62	7,000
63.67	58.77	54.10	45.36	12,886	12,342	11,826	10,888	26.33	26.23	25.90	24.64	8,000
65.50	60.60	55.89	47.06	13,993	13,448	12,929	11,981	24.50	24.40	24.11	22.94	9,000
67.08	62.17	57.45	48.53	15,085	14,540	14,018	13,061	22.92	22.83	22.55	21.47	10,000
72.64	67.71	62.91	53.72	20,405	19,856	19,327	18,339	17.36	17.29	17.09	16.28	15,000
76.01	71.06	66.22	56.87	25,595	25,046	24,512	23,507	13.99	13.94	13.78	13.13	20,000
76.11	71.16	66.32	56.96	25,785	25,235	24,701	23,696	13.89	13.84	13.68	13.04	20,184*
78.27	73.32	68.45	58.99	30,723	30,172	29,635	28,619	11.73	11.68	11.55	11.01	25,000
79.90	74.94	70.06	60.52	35,815	35,263	34,724	33,700	10.10	10.06	9.94	9.48	30,000
81.30	76.33	71.43	61.83	41,679	41,127	40,586	39,555	8.70	8.67	8.57	8.17	35,786†
89.05	84.05	79.06	69.11	384,348	383,793	383,242	382,173	0.95	0.95	0.94	0.89	378,022‡

Table I-1. Earth Satellite Parameters. Continued

	25	26	27	28	29	30	31	32	33	34	35	36	37	38
	EULER ROTATION RATE (deg/min)							EULER AXIS CO–LATITUDE (deg)						
Alt. (km)	Inc = 0.0	Inc = 30.0	Inc = 60.0	Inc = 90.0	Inc = 120.0	Inc = 150.0	Inc = 180.0	Inc = 0.0	Inc = 30.0	Inc = 60.0	Inc = 90.0	Inc = 120.0	Inc = 150.0	Inc = 180.0
0	4.010	4.046	4.141	4.268	4.392	4.480	4.511	0.00	31.78	63.01	93.37	122.83	151.60	180.00
100	3.912	3.948	4.043	4.170	4.293	4.382	4.413	0.00	31.82	63.08	93.45	122.90	151.64	180.00
150	3.864	3.900	3.995	4.123	4.246	4.334	4.365	0.00	31.84	63.12	93.49	122.93	151.66	180.00
175	3.841	3.876	3.972	4.099	4.222	4.310	4.342	0.00	31.85	63.13	93.51	122.95	151.67	180.00
200	3.817	3.853	3.949	4.076	4.199	4.287	4.319	0.00	31.86	63.15	93.53	122.96	151.68	180.00
225	3.794	3.830	3.926	4.053	4.176	4.264	4.295	0.00	31.88	63.17	93.55	122.98	151.69	180.00
250	3.771	3.807	3.903	4.030	4.153	4.241	4.272	0.00	31.89	63.19	93.57	123.00	151.69	180.00
275	3.749	3.784	3.880	4.007	4.131	4.218	4.250	0.00	31.90	63.21	93.58	123.01	151.70	180.00
300	3.726	3.762	3.858	3.985	4.108	4.196	4.228	0.00	31.91	63.23	93.61	123.03	151.71	180.00
325	3.704	3.740	3.836	3.963	4.086	4.174	4.205	0.00	31.92	63.25	93.63	123.05	151.72	180.00
350	3.682	3.718	3.814	3.941	4.064	4.152	4.183	0.00	31.93	63.26	93.65	123.06	151.73	180.00
375	3.660	3.696	3.792	3.919	4.042	4.130	4.161	0.00	31.94	63.28	93.67	123.08	151.74	180.00
400	3.639	3.674	3.770	3.897	4.021	4.108	4.139	0.00	31.96	63.30	93.69	123.10	151.75	180.00
450	3.596	3.632	3.728	3.855	3.978	4.066	4.098	0.00	31.98	63.34	93.73	123.13	151.77	180.00
500	3.554	3.590	3.686	3.813	3.936	4.024	4.056	0.00	32.00	63.38	93.77	123.16	151.79	180.00
550	3.513	3.549	3.645	3.772	3.895	3.983	4.014	0.00	32.02	63.42	93.81	123.20	151.80	180.00
600	3.473	3.508	3.605	3.732	3.855	3.942	3.973	0.00	32.05	63.45	93.85	123.23	151.82	180.00
650	3.433	3.469	3.565	3.692	3.815	3.903	3.935	0.00	32.07	63.49	93.89	123.26	151.84	180.00
700	3.394	3.430	3.526	3.653	3.776	3.864	3.895	0.00	32.09	63.53	93.94	123.30	151.86	180.00
750	3.356	3.392	3.488	3.615	3.738	3.826	3.857	0.00	32.12	63.57	93.98	123.33	151.88	180.00
800	3.318	3.354	3.450	3.578	3.701	3.788	3.819	0.00	32.14	63.61	94.02	123.36	151.90	180.00
850	3.281	3.317	3.413	3.541	3.664	3.751	3.783	0.00	32.17	63.65	94.06	123.40	151.92	180.00
900	3.245	3.281	3.377	3.505	3.627	3.715	3.746	0.00	32.19	63.69	94.10	123.43	151.93	180.00
950	3.209	3.245	3.342	3.469	3.592	3.679	3.710	0.00	32.21	63.73	94.14	123.47	151.95	180.00
1,000	3.174	3.210	3.306	3.434	3.557	3.644	3.674	0.00	32.24	63.77	94.19	123.50	151.97	180.00
1,100	3.106	3.142	3.238	3.366	3.488	3.576	3.607	0.00	32.29	63.84	94.27	123.57	152.01	180.00
1,200	3.039	3.075	3.172	3.300	3.422	3.509	3.541	0.00	32.34	63.92	94.36	123.64	152.05	180.00
1,300	2.975	3.011	3.108	3.236	3.358	3.445	3.476	0.00	32.39	64.01	94.44	123.71	152.09	180.00
1,400	2.913	2.950	3.046	3.174	3.296	3.383	3.415	0.00	32.44	64.09	94.53	123.78	152.12	180.00
1,500	2.853	2.890	2.986	3.114	3.237	3.323	3.354	0.00	32.49	64.17	94.62	123.85	152.16	180.00
2,000	2.580	2.616	2.714	2.841	2.964	3.050	3.081	0.00	32.75	64.59	95.06	124.20	152.36	180.00
2,500	2.344	2.381	2.479	2.607	2.729	2.814	2.845	0.00	33.02	65.03	95.52	124.56	152.55	180.00
3,000	2.139	2.176	2.275	2.403	2.525	2.610	2.640	0.00	33.30	65.48	95.99	124.93	152.75	180.00
3,500	1.960	1.998	2.097	2.225	2.346	2.431	2.461	0.00	33.60	65.94	96.47	125.31	152.96	180.00
4,000	1.802	1.840	1.940	2.068	2.189	2.273	2.303	0.00	33.91	66.43	96.96	125.69	153.16	180.00
4,500	1.662	1.701	1.801	1.929	2.050	2.134	2.163	0.00	34.23	66.92	97.47	126.08	153.37	180.00
5,000	1.538	1.576	1.677	1.806	1.926	2.009	2.039	0.00	34.56	67.44	97.98	126.47	153.58	180.00
6,000	1.325	1.365	1.467	1.596	1.715	1.797	1.826	0.00	35.27	68.51	99.04	127.27	154.00	180.00
7,000	1.152	1.192	1.296	1.425	1.543	1.625	1.653	0.00	36.04	69.64	100.13	128.09	154.42	180.00
8,000	1.008	1.049	1.154	1.284	1.401	1.481	1.510	0.00	36.86	70.84	101.26	128.91	154.85	180.00
9,000	0.887	0.930	1.036	1.165	1.282	1.361	1.389	0.00	37.75	72.10	102.42	129.75	155.28	180.00
10,000	0.785	0.828	0.936	1.065	1.181	1.259	1.286	0.00	38.71	73.42	103.61	130.59	155.71	180.00
15,000	0.444	0.493	0.609	0.738	0.848	0.920	0.945	0.00	44.72	80.88	109.85	134.83	157.83	180.00
20,000	0.256	0.315	0.439	0.565	0.668	0.734	0.757	0.00	53.41	89.66	116.33	138.96	159.83	180.00
20,184*	0.251	0.311	0.434	0.561	0.663	0.729	0.752	0.00	53.79	90.00	116.57	139.11	159.90	180.00
25,000	0.140	0.214	0.343	0.464	0.560	0.620	0.641	0.00	65.86	99.31	122.70	142.83	161.66	180.00
30,000	0.062	0.158	0.287	0.401	0.489	0.545	0.564	0.00	82.63	109.19	128.71	146.36	163.31	180.00
35,786†	0.000	0.130	0.251	0.355	0.434	0.484	0.501	0.00	105.00	120.00	135.00	150.00	165.00	180.00
378,022‡	0.240	0.243	0.246	0.251	0.255	0.259	0.261	0.00	178.93	178.17	177.92	178.23	178.99	180.00

*GPS Altitude †GEO ‡Moon and L4, L5 Lagrange Points

See front of table for formulas and sources. See the book website for an Excel version of this table that allows you to plot the variables vs. altitude, to adjust the altitude range under consideration, and to determine these parameters for orbits around other celestrial bodies.

Table I-1. Earth Satellite Parameters. Continued

39	40	41	42	43	44	45	46	47	48	49	
	ATMOSPHERIC SCALE HEIGHT (km)			ATMOSPHERIC DENSITY			ΔV TO MAINTAIN ALTITUDE				
Inertial Rot. Rate (deg/min)	Solar Min	Solar Mean	Solar Max	Minimum (kg/m^3)	Mean (kg/m^3)	Maximum (kg/m^3)	Solar Min 50 kg/m^2 (m/s)/yr	Solar Max 50 kg/m^2 (m/s)/yr	Solar Min 200 kg/m^2 (m/s)/yr	Solar Max 200 kg/m^2 (m/s)/yr	Alt. (km)
4.261	8.6	8.6	8.6	1.20×10^{0}	1.20×10^{0}	1.20×10^{0}	2.37×10^{13}	2.37×10^{13}	5.93×10^{12}	5.93×10^{12}	0
4.163	5.8	5.8	5.8	5.71×10^{-7}	5.69×10^{-7}	5.67×10^{-7}	1.11×10^{7}	1.10×10^{7}	2.77×10^{6}	2.75×10^{6}	100
4.115	21.7	24.6	27.9	1.90×10^{-9}	2.02×10^{-9}	2.21×10^{-9}	3.65×10^{4}	4.25×10^{4}	9.12×10^{3}	1.06×10^{4}	150
4.091	26.5	29.9	34.8	6.42×10^{-10}	7.66×10^{-10}	9.21×10^{-10}	1.23×10^{4}	1.77×10^{4}	3.08×10^{3}	4.41×10^{3}	175
4.068	29.4	35.3	41.6	2.18×10^{-10}	2.90×10^{-10}	3.84×10^{-10}	4.16×10^{3}	7.34×10^{3}	1.04×10^{3}	1.83×10^{3}	200
4.045	31.7	38.4	45.5	9.64×10^{-11}	1.46×10^{-10}	2.12×10^{-10}	1.84×10^{3}	4.04×10^{3}	4.59×10^{2}	1.01×10^{3}	225
4.022	34.1	41.4	49.4	4.27×10^{-11}	7.30×10^{-11}	1.17×10^{-10}	8.10×10^{2}	2.22×10^{3}	2.02×10^{2}	5.55×10^{2}	250
3.999	35.9	43.6	51.9	2.14×10^{-11}	4.10×10^{-11}	7.17×10^{-11}	4.05×10^{2}	1.35×10^{3}	1.01×10^{2}	3.39×10^{2}	275
3.977	37.7	45.8	54.5	1.07×10^{-11}	2.30×10^{-11}	4.39×10^{-11}	2.02×10^{2}	8.27×10^{2}	5.05×10^{1}	2.07×10^{2}	300
3.955	39.1	47.4	56.4	5.83×10^{-12}	1.38×10^{-11}	2.85×10^{-11}	1.09×10^{2}	5.35×10^{2}	2.73×10^{1}	1.34×10^{2}	325
3.933	40.5	49.0	58.3	3.17×10^{-12}	8.33×10^{-12}	1.85×10^{-11}	5.92×10^{1}	3.46×10^{2}	1.48×10^{1}	8.65×10^{1}	350
3.911	42.0	50.3	59.7	1.81×10^{-12}	5.24×10^{-12}	1.25×10^{-11}	3.38×10^{1}	2.32×10^{2}	8.44×10^{0}	5.81×10^{1}	375
3.889	43.5	51.7	61.1	1.04×10^{-12}	3.29×10^{-12}	8.43×10^{-12}	1.92×10^{1}	1.56×10^{2}	4.81×10^{0}	3.91×10^{1}	400
3.847	48.1	54.2	63.5	3.68×10^{-13}	1.39×10^{-12}	4.05×10^{-12}	6.77×10^{0}	7.46×10^{1}	1.69×10^{0}	1.87×10^{1}	450
3.805	56.6	57.3	65.6	1.40×10^{-13}	6.15×10^{-13}	2.03×10^{-12}	2.56×10^{0}	3.72×10^{1}	6.40×10^{-1}	9.29×10^{0}	500
3.764	71.1	62.0	67.9	5.76×10^{-14}	2.84×10^{-13}	1.05×10^{-12}	1.05×10^{0}	1.91×10^{1}	2.61×10^{-1}	4.79×10^{0}	550
3.723	92.1	69.3	70.7	2.61×10^{-14}	1.37×10^{-13}	5.63×10^{-13}	4.70×10^{-1}	1.01×10^{1}	1.17×10^{-1}	2.54×10^{0}	600
3.684	116.8	80.5	74.7	1.32×10^{-14}	6.87×10^{-14}	3.08×10^{-13}	2.36×10^{-1}	5.51×10^{0}	5.91×10^{-2}	1.38×10^{0}	650
3.645	141.3	95.9	80.5	7.55×10^{-15}	3.63×10^{-14}	1.73×10^{-13}	1.34×10^{-1}	3.07×10^{0}	3.35×10^{-2}	7.68×10^{-1}	700
3.606	163.7	114.6	88.6	4.81×10^{-15}	2.02×10^{-14}	9.95×10^{-14}	8.48×10^{-2}	1.75×10^{0}	2.12×10^{-2}	4.39×10^{-1}	750
3.569	183.87	134.80	99.50	3.34×10^{-15}	1.21×10^{-14}	5.88×10^{-14}	5.86×10^{-2}	1.03×10^{0}	1.46×10^{-2}	2.57×10^{-1}	800
3.532	202.60	154.77	113.09	2.47×10^{-15}	7.69×10^{-15}	3.57×10^{-14}	4.30×10^{-2}	6.22×10^{-1}	1.07×10^{-2}	1.55×10^{-1}	850
3.496	220.77	173.43	128.76	1.90×10^{-15}	5.24×10^{-15}	2.25×10^{-14}	3.28×10^{-2}	3.88×10^{-1}	8.20×10^{-3}	9.70×10^{-2}	900
3.460	238.66	190.33	145.55	1.50×10^{-15}	3.78×10^{-15}	1.46×10^{-14}	2.57×10^{-2}	2.51×10^{-1}	6.47×10^{-3}	6.28×10^{-2}	950
3.425	256.19	205.52	162.51	1.20×10^{-15}	2.86×10^{-15}	9.91×10^{-15}	2.05×10^{-2}	1.69×10^{-1}	5.12×10^{-3}	4.22×10^{-2}	1,000
3.356	—	—	—	—	—	—	—	—	—	—	1,100
3.290	—	—	—	—	—	—	—	—	—	—	1,200
3.226	—	—	—	—	—	—	—	—	—	—	1,300
3.164	—	—	—	—	—	—	—	—	—	—	1,400
3.104	—	—	—	—	—	—	—	—	—	—	1,500
2.830	—	—	—	—	—	—	—	—	—	—	2,000
2.595	—	—	—	—	—	—	—	—	—	—	2,500
2.390	—	—	—	—	—	—	—	—	—	—	3,000
2.211	—	—	—	—	—	—	—	—	—	—	3,500
2.053	—	—	—	—	—	—	—	—	—	—	4,000
1.913	—	—	—	—	—	—	—	—	—	—	4,500
1.788	—	—	—	—	—	—	—	—	—	—	5,000
1.576	—	—	—	—	—	—	—	—	—	—	6,000
1.403	—	—	—	—	—	—	—	—	—	—	7,000
1.259	—	—	—	—	—	—	—	—	—	—	8,000
1.138	—	—	—	—	—	—	—	—	—	—	9,000
1.035	—	—	—	—	—	—	—	—	—	—	10,000
0.694	—	—	—	—	—	—	—	—	—	—	15,000
0.507	—	—	—	—	—	—	—	—	—	—	20,000
0.501	—	—	—	—	—	—	—	—	—	—	20,184*
0.390	—	—	—	—	—	—	—	—	—	—	25,000
0.313	—	—	—	—	—	—	—	—	—	—	30,000
0.251	—	—	—	—	—	—	—	—	—	—	35,786†
0.009	—	—	—	—	—	—	—	—	—	—	378,022‡

Table I-1. Earth Satellite Parameters. Continued

	50	51	52	53	54	55	56	57	58	59	60	61
	ORBIT DECAY RATE				ESTIMATED ORBIT LIFETIME				VELOCITY–RELATED PARAMETERS			
Alt. (km)	Solar Min 50 kg/m² (km/yr)	Solar Max 50 kg/m² (km/yr)	Solar Min 200 kg/m² (km/yr)	Solar Max 200 kg/m² (km/yr)	Solar Min 50 kg/m² (days)	Solar Max 50 kg/m² (days)	Solar Min 200 kg/m² (days)	Solar Max 200 kg/m² (days)	Plane Change ΔV (m/s)/deg	ΔV Req'd for a 1 km Alt Chg (m/s)	Max Ang Rate Gnd Stn (deg/s)	Sun Synchronous Inclination (deg)
0	3.83×10^{13}	3.83×10^{13}	9.57×10^{12}	9.58×10^{12}	—	—	—	—	137.97	0.62	—	
100	1.83×10^{7}	1.82×10^{7}	4.58×10^{6}	4.54×10^{6}	0.06	0.06	0.06	0.06	136.90	0.61	4.49	95.68
150	6.10×10^{4}	7.10×10^{4}	1.52×10^{4}	1.78×10^{4}	0.23	0.24	0.57	0.49	136.38	0.60	2.98	96.00
175	2.07×10^{4}	2.97×10^{4}	5.18×10^{3}	7.42×10^{3}	0.67	0.54	1.90	1.43	136.12	0.59	2.55	96.16
200	7.03×10^{3}	1.24×10^{4}	1.76×10^{3}	3.10×10^{3}	1.97	1.23	6.34	4.19	135.86	0.59	2.23	96.24
225	3.12×10^{3}	6.86×10^{3}	7.80×10^{2}	1.71×10^{3}	4.73	2.37	16.66	8.56	135.60	0.59	1.98	96.33
250	1.38×10^{3}	3.79×10^{3}	3.46×10^{2}	9.49×10^{2}	11.36	4.58	43.77	17.49	135.35	0.58	1.78	96.41
275	6.96×10^{2}	2.33×10^{3}	1.74×10^{2}	5.82×10^{2}	25	8	93	31	135.09	0.58	1.61	96.50
300	3.49×10^{2}	1.43×10^{3}	8.74×10^{1}	3.57×10^{2}	55	14	196	54	134.84	0.58	1.48	96.58
325	1.90×10^{2}	9.30×10^{2}	4.75×10^{1}	2.32×10^{2}	103	23	333	83	134.59	0.58	1.36	96.67
350	1.03×10^{2}	6.05×10^{2}	2.59×10^{1}	1.51×10^{2}	190	37	565	128	134.34	0.57	1.26	96.76
375	5.93×10^{1}	4.09×10^{2}	1.48×10^{1}	1.02×10^{2}	312	55	703	182	134.09	0.57	1.17	96.85
400	3.40×10^{1}	2.76×10^{2}	8.51×10^{0}	6.91×10^{1}	511	83	875	258	133.84	0.57	1.10	96.94
450	1.21×10^{1}	1.33×10^{2}	3.02×10^{0}	3.33×10^{1}	741	157	1,206	502	133.35	0.56	0.97	97.03
500	4.62×10^{0}	6.72×10^{1}	1.16×10^{0}	1.68×10^{1}	953	270	1,727	1,106	132.86	0.55	0.87	97.21
550	1.91×10^{0}	3.50×10^{1}	4.78×10^{-1}	8.74×10^{0}	1,224	464	5,887	6,548	132.38	0.55	0.79	97.40
600	8.67×10^{-1}	1.87×10^{1}	2.17×10^{-1}	4.68×10^{0}	1,613	941	13,658	12,979	131.91	0.54	0.72	97.59
650	4.41×10^{-1}	1.03×10^{1}	1.10×10^{-1}	2.57×10^{0}	5,614	6,053	26,181	25,978	131.44	0.54	0.66	97.79
700	2.53×10^{-1}	5.79×10^{0}	6.32×10^{-2}	1.45×10^{0}	12,883	12,355	50,840	52,512	130.97	0.53	0.61	97.99
750	1.62×10^{-1}	3.34×10^{0}	4.04×10^{-2}	8.36×10^{-1}	23,521	24,069	97,916	97,437	130.51	0.52	0.57	98.19
800	1.13×10^{-1}	1.98×10^{0}	2.82×10^{-2}	4.96×10^{-1}	42,097	44,167	1.75×10^{5}	1.76×10^{5}	130.06	0.52	0.53	98.39
850	8.36×10^{-2}	1.21×10^{0}	2.09×10^{-2}	3.03×10^{-1}	74,235	73,585	3.05×10^{5}	3.06×10^{5}	129.61	0.51	0.50	98.60
900	6.45×10^{-2}	7.63×10^{-1}	1.61×10^{-2}	1.91×10^{-1}	1.30×10^{5}	1.30×10^{5}	5.19×10^{5}	5.20×10^{5}	129.16	0.51	0.47	98.82
950	5.10×10^{-2}	4.99×10^{-1}	1.28×10^{-2}	1.25×10^{-1}	2.11×10^{5}	2.13×10^{5}	8.52×10^{5}	8.52×10^{5}	128.72	0.50	0.44	99.03
1,000	4.11×10^{-2}	3.39×10^{-1}	1.03×10^{-2}	8.47×10^{-2}	3.40×10^{5}	3.39×10^{5}	1.36×10^{6}	1.36×10^{6}	128.28	0.50	0.42	99.25
1,100	—	—	—	—	6.46×10^{5}	6.46×10^{5}	2.59×10^{6}	2.59×10^{6}	127.42	0.49	0.38	99.48
1,200	—	—	—	—	1.23×10^{6}	1.23×10^{6}	4.93×10^{6}	4.93×10^{6}	126.58	0.48	0.35	99.94
1,300	—	—	—	—	2.09×10^{6}	2.09×10^{6}	—	—	125.75	0.47	0.32	100.42
1,400	—	—	—	—	3.17×10^{6}	3.17×10^{6}	—	—	124.94	0.46	0.29	100.91
1,500	—	—	—	—	4.81×10^{6}	4.81×10^{6}	—	—	124.14	0.45	0.27	101.43
2,000	—	—	—	—	—	—	—	—	120.38	0.41	0.20	101.96
2,500	—	—	—	—	—	—	—	—	116.94	0.38	0.15	104.89
3,000	—	—	—	—	—	—	—	—	113.78	0.35	0.12	108.35
3,500	—	—	—	—	—	—	—	—	110.87	0.32	0.10	112.41
4,000	—	—	—	—	—	—	—	—	108.16	0.30	0.089	117.21
4,500	—	—	—	—	—	—	—	—	105.65	0.28	0.077	122.93
5,000	—	—	—	—	—	—	—	—	103.30	0.26	0.068	129.86
6,000	—	—	—	—	—	—	—	—	99.04	0.23	0.054	138.59
7,000	—	—	—	—	—	—	—	—	95.27	0.20	0.045	—
8,000	—	—	—	—	—	—	—	—	91.89	0.18	0.038	—
9,000	—	—	—	—	—	—	—	—	88.86	0.17	0.032	—
10,000	—	—	—	—	—	—	—	—	86.10	0.15	0.028	—
15,000	—	—	—	—	—	—	—	—	75.36	0.10	0.016	—
20,000	—	—	—	—	—	—	—	—	67.85	0.074	0.011	—
20,184*	—	—	—	—	—	—	—	—	67.61	0.073	0.011	—
25,000	—	—	—	—	—	—	—	—	62.21	0.057	0.008	—
30,000	—	—	—	—	—	—	—	—	57.77	0.045	0.006	—
35,786†	—	—	—	—	—	—	—	—	53.66	0.036	0.005	—
378,022‡	—	—	—	—	—	—	—	—	17.77	0.001	0.000	—

*GPS Altitude †GEO ‡Moon and L4, L5 Lagrange Points

See front of table for formulas and sources. See the book website for an Excel version of this table that allows you to plot the variables vs. altitude, to adjust the altitude range under consideration, and to determine these parameters for orbits around other celestrial bodies.

5 Mission Analysis and Mission Utility

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5.1 Introduction to Mission Analysis
The Mission Analysis Hierarchy

5.2 Studies with Limited Scope
Feasibility Assessment; Sizing Estimate; Point Design

5.3 System Trade Studies and Performance Assessments (Step 8)
System Trade Studies; Performance Assessments

5.4 Mission Utility and Figures of Merit—Is the Mission Worthwhile? (Step 9)
Performance Parameters and Measures of Effectiveness; Evaluating Mission Utility

5.5 Defining the Baseline Mission Concept, Revising Requirements, and Evaluating Alternatives (Steps 10–12)

5.6 Examples: FireSat II and SCS

5.7 Deciding Whether a Mission Should Proceed

Chapter 4 discussed defining and characterizing alternative concepts and architectures for space missions. This chapter shows how we evaluate the ability of these options to meet fundamental mission objectives. We address *mission analysis*—how to quantify mission performance, *mission utility*—how to evaluate its utility to the customer or end user, and *mission selection*—how to select one or more concepts for further development or to decide that we cannot achieve the mission within current constraints or technology.

The narrow purpose of mission evaluation is to determine whether the system meets all of the requirements and specifications set for it. This is critically important (particularly to Congress or a group of investors who are putting up the large amounts of money that space missions normally require) and is largely the subject of Chap. 6. However, in the broad context of mission engineering, we are really concerned with two issues:

1. How well does the system perform (*performance analysis*)
2. How well does it meet the customer's or end user's objectives (*mission utility analysis*)

The distinction between performance and utility can be exceptionally important. This is clearly illustrated by two historical examples—the Space Shuttle and the Iridium LEO Communications Constellation.

The original goal of the Space Shuttle was to achieve "airplane-like" access to space and to drive down the launch cost to less than $10 million/flight in FY1972$, corresponding to roughly $50 million/launch in FY2010$. The actual cost per launch is not well-defined, but is likely well over $1 billion/launch, not counting any amortization of the original development cost. Irrespective of the exact costs, it is clear that the Shuttle was never able to achieve the dramatic cost reductions that were promised. As discussed in Sec. 2.1.7 and Chap. 13, high launch costs remain a major impediment to the large-scale expansion of the exploration and use of space.

Similarly, the Iridium LEO communications constellation declared bankruptcy in 2000, representing a loss of approximately $5.5 billion—most likely from a group of people that would have preferred to not lose that money. As discussed in Sec. 2.1.4, this represented a loss for the astronautics community as a whole. It indirectly increased costs for nearly all other programs and made it difficult or impossible to fund other constellations.

The important issue is that both Iridium and the Space Shuttle largely met their performance requirements.* (The Shuttle program had two failures resulting in loss of life, which represents a major and significant programmatic failure, even though we know that human spaceflight carries risks that can not realistically be avoided.) Both programs were exceptionally well-engineered, and were built by some of the best engineers and space organizations in the world. Nonetheless, while the performance assessment was very good, they failed to meet their fundamental mission objectives, i.e., to achieve an adequate level of mission utility. We, as a community, did not do what we should have done. If a surgeon performs a remarkable operation, but the patient dies, they should not take comfort in the great work that

* For the Space Shuttle the performance requirements are things like payload mass to orbit, injection accuracy, and loiter time on orbit, whereas the mission objective was to achieve "airplane-like" performance and dramatically drive down launch costs. For Iridium the performance requirements were availability, number of telephone calls supported, and ability to successfully complete a call. The objective was simple—to make money for the people who put up the initial investment.

they did. The only immediate goal of the surgery is to help the patient. If Iridium goes bankrupt or the Shuttle fails to drive down launch cost, then we didn't do a good job*. This is the fundamental purpose of mission utility analysis—to evaluate and quantify what it means to meet the needs of the end user.

Essentially all missions go through mission analysis and evaluation stages many times. The literature base describing this process has increased significantly in recent years. Fortescue et al. [2003], Pisacane and Moore [2005], Larson et al. [2009], and Chesley et al. [2008] discuss the process for generic missions; Larson and Pranke [1999] do so for human missions; Przemieniecki [1993, 2000] does this for defense missions; Kemble [2006] does so for interplanetary missions; and NASA [2007] provides an excellent process summary for NASA missions. Kay [1995] discusses the difficulty of doing this within the framework of a political democracy and Sarsfield [1998], Sandau [2006], and Wertz and Larson [1996] provide specific techniques applicable to reducing mission cost, which is also discussed in Chap. 13.

5.1 Introduction to Mission Analysis

Mission analysis is the process of quantifying the system parameters and resulting performance. *Mission utility analysis*, described in Sec. 5.4, is the process of quantifying how well the system meets its overall mission objectives. Recall that the mission objectives themselves are not quantitative. However, our capacity to meet them should be quantified as well as possible in order to allow us to make intelligent decisions about whether and how to proceed. *Mission requirements,* introduced in Chap. 3 and discussed in more detail in Chap. 6, are the numerical expressions of how well the objectives must be met. At their best, they represent a balance between what we want and what is feasible within the constraints on the system and, therefore, deriving them should be a central part of the mission analysis activity. In practice, requirements are often set by a process outside of the space mission and unrelated to what is real or achievable. Therefore, it is often the case that mission analysis is concerned with how and how well externally defined mission requirements can be met. Nonetheless, in principle, mission analysis should be the process by which we define and refine both the system definition and mission requirements in order to meet our broad objectives at minimum cost and risk.

A key component of mission analysis is documentation, which provides the organizational memory of both the results and reasons. It is critical to understand fully the choices made, even those that are neither technical nor optimal. We may choose to apply a particular technology for political or economic reasons, or may not have enough manpower to investigate alternatives. In any case, for successful analysis, we must document the real reasons so others can reevaluate them later when the situation may be different. Technical people often shy away from nontechnical reasons or try to justify decisions by exaggerating their technical content. For example, as a result of our preliminary FireSat II analysis, we may choose a circular orbit at 700 km at an inclination of 60 deg because this is a good mid-range starting point. If so, we should document this as the reason, rather than trying to further justify these parameters or simply leave the reason undefined. If we have documented the reason, then later we, or others, can refine the choice for the best altitude and inclination, rather than having to live by choices for which there is no documented justification.

5.1.1 The Mission Analysis Hierarchy

I like to think of the mission analysis process as a huge spreadsheet model of a space system. Along the top of the spreadsheet matrix are the various parameters and alternatives that one might assess, such as power, orbit, and staffing levels for ground stations. Along the left are the various system parameters that flow from the mission choices, such as the orbit period, or coverage parameters, and along the bottom row are the system's quantitative outputs, indicating its performance, effectiveness, cost, and risk. The matrix itself would capture the functional relationships among the many variables. We would like to wiggle any particular parameter—such as the altitude, or the number of people assigned to the ground station—and determine the effect on all of the other parameters. In this way, we could quantify the system's performance as a function of all possible variables and their combinations.

Fortunately for the continuing employment of mission analysts, the above spreadsheet model does not yet exist.† Instead, we analyze as many reasonable alternatives as possible so we may understand how the system

* What does "we" mean in this context? Obviously most aerospace engineers weren't directly involved in either Iridium or the Space Shuttle. Nonetheless, all of us collectively set the rules and define the "culture" in which we all work. It is this culture that has let it be acceptable in some sense, for the Shuttle to miss its recurring cost projections by a factor of 20 or so and for Iridium to go bankrupt, rather than make a good return on investment for those that put up the money. Our surgeon analogy continues to work here. Surgeons, collectively aren't responsible for the death of a single patient. But they do make the rules under which surgeons operate. If that death causes a reassessment of those rules, it's still not good, but at least the community is responding. The fundamental question for mission utility and space mission engineering is whether our community is responding appropriately and adequately to the fact that programs cost too much and take too long.

† In due course, system engineers may become technologically obsolete. The challenge to future system engineers will be to stay ahead of the computer models in being creative and innovative. One of the elements most often missed by automated processes is how to think about the problem in a totally different way. For example, can we drive down the cost of a human mission to Jupiter by first putting a settlement on the Moon and using those resources to build low-cost interplanetary systems? These are the types of questions that are very difficult to answer by simply adjusting parameters in a model.

behaves as a function of the principal design features, i.e., the system drivers (Chap. 4). This approach does not imply that we are uninterested in secondary detail, but simply recognizes that the mission analysis process, much like the space system we are attempting to analyze, is ultimately limited in both cost and schedule. We must achieve the maximum level of understanding within these limits.

If the resources available for concept exploration are limited, as is nearly always the case in realistic situations, then one of the most critical tasks is to intelligently limit the scope of individual analyses. We want to be able to compute approximate values for many parameters and to determine at what level of detail we should reasonably stop. In practice, this is made difficult by the continuing demand for additional detail and depth. Thus, we would like to be able to determine and make clear to others what elements of that detail will significantly affect the overall system performance and what elements, while important, can reasonably be left to a more detailed design phase.

We use two main methods to limit the depth of analysis in any particular area. The first is to clearly identify each area's system drivers by the methods in Sec. 4.4 and to concentrate most of the mission analysis effort on these drivers. The second is to clearly identify the goal of the system study and to provide a level of detail appropriate to that goal. This approach leads to a *mission analysis hierarchy*, summarized in Table 5-1, in which studies take on increased levels of detail and complexity as the activity progresses. The first three types of studies are meant to be quick with limited detail and are not intended to provide definitive results. The last three are much more complex ways to select an alternative to provide the best system performance. Each of these types of analysis is discussed below.

5.2 Studies with Limited Scope

The first three types of analyses in Table 5-1 provide methods for undertaking a *quick-look assessment*. They provide limited detail, but can frequently be done quickly and at low cost. Consequently, these quick-look assessments are important in any situation that is funding-limited or time-limited, which is very often the case in real mission analysis. We will briefly outline these methods here. However, nearly the entire book is devoted to the process of making initial estimates, which is the basic goal of limited scope studies. We want to be able to understand whether or not a particular project is feasible, and to get some idea of its size, complexity, and cost. Doing this requires that we be able to make numerical estimates and undertake limited studies in order to develop insight into the nature of the problem we are trying to solve.

The biggest difficulty with limited scope studies is the tendency to believe that they are more accurate than they really are. Thus, it is not uncommon to use a feasibility assessment or point design to establish the requirements for a mission in such detail that, in practice, the point design becomes the only alternative which can meet them. As long as we recognize the limited scope of these studies, they have a valuable place in the mission analysis hierarchy and represent one of the most important tools that we can use to understand the behavior of the system we are designing.

5.2.1 Feasibility Assessment

The simplest procedure in the mission analysis hierarchy is the *feasibility assessment*, which we use to establish whether a particular objective is achievable and to place broad bounds on its level of complexity. Frequently, we can do a feasibility assessment simply by comparison with existing systems. Thus, we are reasonably convinced that FireSat II is feasible because most FireSat II tasks could be performed by existing Earth resources satellites. Similarly, it is feasible to land people on the Moon and return them safely to Earth because we have done so in the past.

We can also determine whether a particular goal is feasible by extrapolating our past experience. Is it feasible to send people to Mars and bring them back safely? Here we need to look at the principal differences

Table 5-1. The Mission Analysis Hierarchy. During the preliminary design phase, these help us decide how detailed studies should be.

Analysis Type	Where Discussed	Goal	
Feasibility Assessment	5.2.1	To establish whether an objective is achievable and its approximate degree of complexity	Quick, limited detail
Sizing Estimate	5.2.2	To estimate basic parameters such as size, weight, power, and cost	
Point Design	5.2.3	To demonstrate feasibility and establish a baseline for comparison of alternatives	
Trade Study	5.3.1	To establish the relative advantages of alternative approaches or options	More detailed, complex trades
Performance Assessment	5.3.2	To quantify performance parameters (e.g., resolution, timeliness) for a particular approach	
Utility Assessment	5.4	To quantify how well the system meets overall mission objectives and satisfies the needs of the end user	

between a Mars mission and a Lunar mission. These differences include a longer flight time and higher gravity at Mars than on the Moon, and, therefore, a need for greater radiation protection during the mission and a higher lift-off velocity to leave Mars. These factors make the job more challenging and possibly more expensive than going to the Moon, but there is nothing about the Mars mission that makes it inherently impossible. Getting to Mars is feasible. The problem is being able to do so at modest cost and risk such that we are willing to both pay for the mission and accept the risk.

The third method of providing a feasibility assessment is to provide a very broad design of how such a mission might be accomplished. For example, in the 1970s, Gerard O'Neill of Princeton University proposed building large space colonies at the Lagrange points between the Earth and the Moon. (See Sec. 2.1.9 and O'Neill [1974].) No mission of this scope had ever been undertaken (and still hasn't), and it certainly was not a straightforward extrapolation of any of our normal space experience. O'Neill and his colleagues proceeded to establish the feasibility by developing a variety of alternative designs for such space colonies. While the work done was far in excess of a simple feasibility assessment, it clearly established that such colonies were feasible and gave at least an estimate of the scope of the problem.

5.2.2 Sizing Estimate

The purpose of the *sizing estimate* is to provide an estimate of basic mission parameters such as size, weight, power, or cost. We can do sizing estimates in much the same manner as the feasibility assessment: by analogy with existing systems. Thus, if we are aware of an Earth observation system that has resolution and information characteristics comparable to what we believe are needed for FireSat II, we can use these parameters to give us an initial estimate of the FireSat II parameters.

We can also provide a quantitative estimate of key mission parameters by scaling the parameters from existing missions or payloads in order to obtain estimates of the component sizes for our particular mission. This scaling process is described in Sec. 17.2.6 for space payloads, and in Sec. 14.5 for the spacecraft as a whole. The process of sizing by scaling existing hardware is an extremely powerful approach to estimating what it will take to achieve mission objectives. It is of use not only during the conceptual design process, but throughout the hardware design, definition, and development phases to evaluate the system design as it evolves. If scaling existing systems leads to the suggestion that a particular component should be twice as heavy as the current design suggests, this gives us reason to look very closely at the current design and to try to determine whether or not any factors have been overlooked. We assume that designers of previous systems did a reasonable job of optimizing their system. If the current design is significantly different, either better or worse, then we would like to understand the reasons for these differences. It may be that the technology has advanced to the point where a better design is possible or it may be that the current preliminary design has simply forgotten a key element. This process is a good way to gain confidence in the overall design as we proceed.

As the design develops, more and more accurate sizing estimates come from the scaling process. We proceed by breaking down the system into components and sizing individual components based on scaling estimates with prior systems. Thus, we may initially estimate the system as a whole divided into a spacecraft and payload. As the design becomes more detailed, we will break down the spacecraft into its subsystems and components and estimate the size, weight, and power of each element based upon scaling from prior systems or engineering estimates of the new system to be built. Similarly, we initially size the payload or ground station by comparison with existing systems and eventually by building a list of all the components and undertaking similar sizing estimates for each component. As introduced in Chap. 3, this process of creating a list of components and estimating parameters for each is known as budgeting and is described in more detail in Sec. 14.5.

5.2.3 Point Design

A *point design* is a design, possibly at a top level, for the entire system that is capable of meeting the broad mission objectives. We refer to it as a point design if we have not attempted to optimize the design to either maximize performance or minimize weight, cost, or risk. The point design serves two basic purposes. It demonstrates that the mission is feasible, and it can be used as a baseline for comparison of alternatives. Thus, if we can establish a point design for FireSat II that meets mission objectives with a spacecraft that weighs 500 kg and costs $50 million, then we can use this as a comparison for later systems. If other systems cost more, weigh more, and don't perform as well, then we will abandon those alternatives in favor of the original baseline. If we continue to improve the design so that the cost and risk decrease, then we will let the baseline evolve to take into account the new approaches.

A point design is valuable because we can do it quickly and easily. There is no need to optimize any of the parameters associated with the design unless it is necessary to do so to meet mission objectives. This gives us a sense of whether it will be easy or hard to meet the mission objectives and what are likely to be the most difficult aspects. One of the biggest problems in a point design is taking it too seriously at a later stage. We are always likely to regard work that we have done as representing the best approach, even though we may not have been aware of alternatives. The key issue here is to make use of point designs but at the same time to recognize their limitations and to continue to do trades to reduce overall cost and risk and to look for alternative approaches to meet mission objectives better and at lower cost.

5.3 System Trade Studies and Performance Assessments (Step 8)

Step 8 of our overall Space Mission Engineering process defined in Table 3-1, is to conduct performance assessments and system trades. Deciding whether to proceed with a mission should be based on a strawman system concept or a point design that shows that the mission objectives can be met within the assigned constraints. Of course, the initial point design is probably not the best solution, and we would ordinarily consider a number of alternatives. This process of looking at alternatives is the *system trade process*. Generally there are three types of systems trades that are done, or should be done:

- Single effect trades (easy to do)
- Multiple effects trades (hard to do)
- New and unique approaches (very hard to do)

The system trade process evaluates different broad concepts to establish their viability and impact on performance, cost, risk, and schedule. We then combine the system trade results with the performance assessment (Sec. 5.3.2) and the mission utility analysis (Sec. 5.4) to provide input for concept selection (Sec. 5.7).

5.3.1 System Trade Studies

System trades consist of analyzing and selecting *system drivers*, i.e., key system parameters defined in Sec. 4.4, which largely determine mission performance, cost, risk, and schedule. We use these parameters to define a mission concept and mission architecture that can then be used for performance analysis and utility analysis as described in Secs. 5.3.2 and 5.4. The key system trades are those that define how the system works and determine its size, cost, and risk. Typically, the key system trades will be in one of the major areas defined in Table 5-2. Here the first question should always be: What are the real objectives of the mission? In addition, it would be wise to review the broad objectives of the community involved, as discussed in Chap. 2. What are the long-term goals and objectives of the community and how will this mission support those goals? For example, space telescopes of all sorts are typically out to find things that have never been seen before, which makes defining the system parameters a bit challenging. For military systems, our goal is ordinarily to provide support to the warfighter or gather intelligence about the enemy.

Table 5-2. Principal Areas for Key System Trades. See Table 5-3 for examples.

Area	Principal Issue	Where Discussed
Mission Objectives	What are we really trying to accomplish?	Sec. 4.1
Critical Requirements	What are the requirements that are hard to meet and, therefore, drive cost, risk, and schedule?	Sec. 4.4
Mission Concept	How do we get the information and data and how do we get the results to the end user?	Sec. 4.1
Subject	What do we want to look at or measure to get the information we want?	Sec. 15.2
Type and Complexity of Payloads	What spectral band do we work in and how do we measure it? How many payloads?	Sec. 15.1
Orbit	A specialized orbit or one of a continuum?	Chap. 10
Number of Spacecraft	A single satellite or a constellation?	Chap. 10

Table 5-3. Examples of Representative Key System Trades. Although these system trades are critical, we can't expect numerically precise answers to our system design problem.

Trade Area	FireSat II	SCS	James Webb Space Telescope	Aircraft Detection
Mission Objectives	Save lives and property or reduce fire fighting costs?	Support the warfighter?	Discover stuff we've never seen? If so, how do we build the telescope?	Traffic control or smuggling interdiction?
Critical Requirements	What's a forest fire? What's the needed resolution?	Small omni ground antenna?	How cold a detector? How large an aperture?	How small a plane do we need to detect and track?
Mission Concept	Identify forest fires on board or on the ground?	Circular orbits with fore and aft fixed crosslinks?	How do we deploy and control the mirror?	Is it a cooperative target? How do we cover the areas of interest?
Subject	Heat, fire, or smoke?	Likely a handheld receiver.	Something we've never seen and don't know.	Commercial planes or small private planes?
Payload Type and Complexity	What wavelength? How many spectral bands?	What band? How many?	How many sensors and instruments?	IR, visible, radar, RF?
Orbit	LEO or GEO	LEO, HEO, MEO, GEO	Lagrange point or closer and cheaper?	LEO, HEO, MEO?
Number of Spacecraft	1 or a constellation?	Probably a constellation	Probably only 1	1 or a constellation?

Table 5-3 shows typical examples of key system-trades for several representative missions. For essentially all missions, specification of the critical requirements will be a key system trade. For the FireSat II mission, the subject is probably the heat from the fire itself and the payload is probably an IR sensor. Thus, the principal system trades are probably the mission objectives, mission concept, the resolution and coverage requirements, the orbit, and whether it's a single satellite or a constellation. For a mission such as the James Webb Space Telescope, there will be only 1 telescope, the orbit is of marginal importance, and the subject is moderately well defined, if only very poorly known. Here the principal trades will be the resolution and pointing requirements, the payload, and the mission concept. Communications satellite systems are normally in geosynchronous orbit with a well-defined concept of operations. Here the only real trade is with the required traffic load, the subject, and the size and complexity of the payload.

Truly innovative approaches—those that really change how we think about a problem—typically involve finding a new option among these key system trades. Motorola's Iridium program and subsequent low-Earth orbit communications constellations represented a new way of thinking about using satellites for communications. These have a very different concept of operations and different orbits from traditional systems. Similarly, Chap. 13 presents an innovative approach to thinking about FireSat II that provides a totally different concept of operations and type of payload. Innovative solutions are never easy to come by. To try to find them, a good place to start is with the key system trade areas given in Tables 5-2 and 5-3.

We cannot normally do system trades in a straightforward numerical fashion. Choosing a different concept of operations, for example, will result in changes in most or all of the mission parameters. Consequently, the fact that Option A requires twice the power of Option B may or may not be critical, depending on the orbit and number of satellites for the two options. We need to look at the system as a whole to understand which is better.

The best approach for key system trades is a utility analysis as described in Sec. 5.4. We use the utility analysis to attempt to quantify our ability to meet mission objectives as a function of cost. We then select the option that fulfills our objectives at the lowest cost and risk. As described in Sec. 5.7, this is still not a straightforward numerical comparison, but does have quantitative components.

The simplest option for system trades is a list of the options and the reasons for retaining or eliminating them. This allows us to consider the merits and demerits at a high level without undertaking time-consuming trades. This also allows our list to be challenged at a later date. We should go back to our key system trades on a regular basis and determine whether our assumptions and conclusions are still valid. It is this process of examination and review that allows us to introduce technical innovation and new ideas. It is a process that must occur if we are to drive down the cost of space systems.

Table 5-3◀, Fig. 5-0, Eq. 5-0

Single Effect Trades

For the purpose of trade studies, system drivers generally divide into two categories—those for which more is better (*single effect drivers*) and those with multiple effects. By far the easier to deal with are the "more is better" single effect drivers, for they simply require us to ask: "What is the cost of achieving more of the commodity in question?" For example, in a space-based radar, added power improves performance but costs more money. Thus, the designer will want to understand how much performance is available for how much power. A second example is coverage. For virtually any Earth-oriented system, including our FireSat II example, more coverage means better performance at higher cost. Increasing coverage ordinarily means adding satellites or, perhaps, increasing a single satellite's coverage by increasing its altitude or the range of its sensors. Therefore, we often do a coverage trade considering performance vs. number of satellites, substituting the latter for cost. Assessing performance as a function of power or coverage may take considerable work, but it is relatively easy to present the data for judging by the funding organization, the users, or other decision makers.

Multiple Effect Trades

System drivers and critical requirements that cause multiple effects demand more complex trade studies. The basic problem here is that pushing parameters one way will improve some characteristics and degrade others. In trades of this type, we are looking for a solution that provides the best mix of results. Examples of such trade studies include instrument design, antenna type, and altitude. Each antenna style will have advantages and disadvantages, so we must trade various possible solutions depending upon the end goals and relative importance of different effects.

In trades with multiple effects, selecting the correct independent parameter for each trade is critical. Consider, for example, selecting either a reflector or a phased-array antenna for a space-based radar [Brookner and Mahoney, 1986]. From the radar equation, we know that a principal performance parameter for a radar is the antenna aperture. All other things being equal, larger antennas will provide much better performance. Thus, for our example, we might choose to compare reflector and phased-array antennas of equal aperture. On this basis, we would choose the phased array because its electronic steering makes it more agile than a reflector antenna, which must be mechanically steered. But our choice becomes more complex when we recognize that weight typically limits large space structures more than size does. Generally, we can build a reflector larger than a phased array for a given weight. Based on weight, a reflector may have considerably more power efficiency and, therefore, be a better radar than a phased-array system. Thus, we would have to trade the better performance of a larger reflector vs. the better agility of a smaller phased array. Depending upon the application, the results may be the same as for an aperture-based trade

Table 5-4. System Trade Process for Parameters with Multiple Effects. The example is the altitude trade for the FireSat II mission. See also Fig. 5-1.

Step	FireSat II Example	Where Discussed
1. Select trade parameter (typically a system driver)	Altitude	Sec. 10.2
2. Identify factors which affect the parameter or are affected by it	Coverage Deployment strategy and coverage evolution Orbit period Time in view Eclipse fraction Response time Number of spacecraft needed Launch capability Resolution Radiation environment Communications Lifetime	Sec. 10.2.1 Sec. 10.6.2 Sec. 9.1.5, App. I Sec. 8.4 Sec. 8.2.4 Sec. 10.2.1.4 Sec. 10.6.1 Secs. 10.3.2, 26.1 Sec. 10.2.2.1 Chap. 7 Secs. 16.2, 21.1 Secs. 3.4, 9.4.4, Chap. 30
3. Assess impact of each factor	Can launch up to 1,800 km Best coverage above 600 km Resolution—lower is better Survivability—higher is better, but not much	—
4. Document and summarize results	Launch Coverage Resolution Survivability	Fig. 5-1
5. Select parameter value and possible range	Altitude = 700 km Range = 600 to 800 km	In text

or reverse. The important point is the critical nature of selecting the proper independent variable in system trades. To do so, we must find the quantities that inherently limit the system being considered. These could be weight, power, level of technology, cost, or manufacturability, depending on the technology and circumstances.

Table 5-4 summarizes the system trade process for parameters with multiple effects. Typically the trade parameter is one of our system drivers. We begin by identifying what performance areas or requirements affect or are affected by the trade parameter. For example, the altitude of the spacecraft will have a key effect on coverage, resolution, and survivability and will be limited by launchability, payload weight, communications, and radiation. We next assess the effect in each of these areas and document and summarize the results, generally **without** trying to create a numerical average of different areas. We use the summary to select the parameter value and a possible range. Although the process is complex and may not have a well-defined answer, it is not necessarily iterative unless we find that the results require fundamental changes in other system parameters.

Altitude trades are perhaps the most common example of a trade in which multiple influences push the parameter in different ways. We would normally like to move the satellite higher to achieve better coverage, better survivability, and easier communications. On the other hand, launchability, resolution, and payload weight tend to drive the satellite lower. The radiation environment dictates specific altitudes we would like to avoid, and the eclipse fraction may or may not play a crucial role in the altitude trade. We must assess each of these effects and summarize all of them to complete our trade study. One possible summary is a numerically weighted average of the various outcomes, such as three times the coverage in square nautical miles per second divided by twice the resolution in furlongs. Although this provides a convenient numerical answer, it does not provide the physical insight or conceptual balance needed for intelligent choices. A better solution is to provide the data on all of the relevant parameters and choose based on inspection rather than numerical weighting.

The FireSat II altitude trade provides an example of this process. For FireSat II, neither survivability nor communications is a key issue, but coverage does push the satellite upward. On the other hand, payload weight and good resolution tend to push the satellite lower. Figure 5-1 shows the results of a hypothetical FireSat II altitude trade. Notice that each parameter has various possible outcomes. Altitudes above or below a certain value may be eliminated, or we may simply prefer a general direction, such as lower altitude providing better resolution. Based on these results, we select a nominal altitude of 700 km for FireSat II and a possible range of 600 to 800 km. This selection is neither numerical nor purely qualitative. We have tried to balance the alternatives sensibly, but not in a way that we can numerically justify.

New and Unique Approaches

By far the most challenging system trades, and also those that have the most potential benefit, are trades that involve new and unique approaches to system design. This is where potential major breakthroughs come about that can dramatically alter the way we do business in

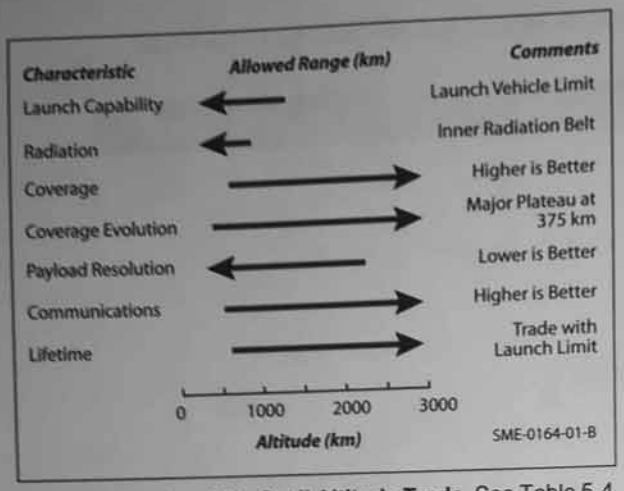


Fig. 5-1. Results of FireSat II Altitude Trade. See Table 5-4 for a list of altitude trade issues. Political constraints and survivability were not of concern for the FireSat II altitude trade.

space. Typically, it's very difficult to make major breakthroughs occur, in part because they may end up being more difficult and less advantageous than was first anticipated (i.e. the reusable Space Shuttle and its goal of dramatically reducing launch cost) and in part because it is difficult in the very conservative space community to move new and innovative approaches beyond the study phase and into real systems (such as the O'Neill space colonies described in Sec. 2.1.9).

An example of this, that also illustrates the potential pitfalls, is the Iridium communications constellation discussed earlier (in the chapter) and in Sec. 2.1.4. Traditionally, communications systems had been thought of as either geosynchronous or Molniya orbits. Iridium took a very different approach and used a LEO constellation instead which, unfortunately, did not work out as intended. On the other hand, both GlobalStar and ORBCOMM are now working successfully in LEO, and Iridium itself has regained sufficient business that an Iridium Next constellation is under development.

A new and unique approach that is working very well is the development of CubeSats, initially intended for educational purposes, but now expanding into highly functional small satellites for a variety of purposes. (See Secs. 2.1.8 and 25.4.)

In the communities that actually fund and build space systems, the acceptance of new approaches moves with all the rapidity of a glacier moving uphill. Nonetheless, we very much need new ideas and new approaches if we are to change the way business is done in space. As we saw in Chap. 1, space missions cost too much and take too long, and we need to find ways to reverse that trend. We will return to this, and discuss explicitly how to do it, in Chap. 13.

5.3.2 Performance Assessments

A *performance assessment* is a numerical analysis of how well the system will perform in specific areas. Computing the area search rate, the available data rate, or the resolution of an optical system are all the result of a performance assessment.

Quantifying performance demands an appropriate level of detail. Too much detail drains resources away from other issues; too little keeps us from determining the important issues or causes us to assess the actual performance incorrectly.

To compute system performance, we use three main techniques:

- System algorithms
- Analogy with existing systems
- Simulation

System algorithms are the basic physical or geometric formulas associated with a particular system or process, such as those for determining resolution in diffraction-limited optics, finding the beam size of an antenna, analyzing a link budget, or assessing geometric coverage. Table 5-5 lists system algorithms typically used for space mission analysis. System algorithms provide the best method for computing performance. They provide clear traceability and establish the relationship between design parameters and performance characteristics. Thus, for FireSat II, we are interested in the resolution of an on-orbit fire detector. Using the formula for diffraction-limited optics in Chap. 17, we can compute the achievable angular resolution from the instrument objective's diameter. We can then apply the geometric formulas in Chap. 8 to translate this angular resolution to resolution on the ground. This result gives us a direct relationship between the altitude of the FireSat II spacecraft, the size of the payload, the angles at which it works, and the resolution with which it can distinguish features on the ground. Similarly, communications systems follow a very-well defined *link budget*, that can determine with

Table 5-5. Common System Algorithms Used for Quantifying Basic Levels of Performance. These analyses use physical or geometrical formulas to determine how system performance varies with key parameters.

Algorithm	Used For	Where Discussed
Link Budget	Communications and data rate analysis	Sec. 16.2
Diffraction-limited Optics	Aperture sizing for optics or antennas; determining resolution	Sec. 17.1
Payload Sensitivity	Payload sizing and performance estimates	Sec. 17.2
Radar Equation	Radar sizing and performance estimates	Cantafio [1989]
Earth Coverage, Area Search Rates	Coverage assessment; system sizing; performance estimates	Chap. 8, App. C
Mapping and Pointing Budget	Geolocation; instrument and antenna pointing; image sensing	Sec. 8.6

good precision, just how fast we will be able to get data back from Jupiter, given the power, wavelength or frequency, and the diameter of the antennas on the ground and on the spacecraft.

System algorithms are powerful in that they show us directly how performance varies with key parameters. However, they are inherently limited because they presume the rest of the system is designed with fundamental physics or geometry as the limiting characteristic. For FireSat II, resolution could also be limited by the optical quality of the lens, by the detector technology, by the spacecraft's pointing stability, or even by the data rates at which the instrument can provide results or that the satellite can transmit to the ground. In using system algorithms, we assume that we have correctly identified what limits system performance. But we must understand that these assumptions may break down as each parameter changes. Finding the limits of these system algorithms helps us analyze the problem and determine its key components. Thus, we may find that a low-cost FireSat II system is limited principally by achieving spacecraft stability at low cost. In this case, our attention would be focused on the attitude control system and on the level of resolution that can be achieved as a function of system cost.

The second method for quantifying performance is by comparing our design with existing systems. In this type of analysis we use the established characteristics of existing sensors, systems, or components and adjust the expected performance according to basic physics or the continuing evolution of technology. The list of payload instruments in Chap. 15 is an excellent starting point for comparing performance with existing systems. We could, for example, use the field of view, resolution, and integration time for an existing sensor and apply them to FireSat II. We then modify the basic sensor parameters such as the aperture, focal length, or pixel size, to satisfy our mission's unique requirements. To do this, we must work with someone who knows the technology, the allowable range of modifications, and their cost. For example, we may be able to improve the resolution by doubling the diameter of the objective, but doing so may cost too much. Thus, to estimate performance based on existing systems, we need information from those who understand the main cost and performance drivers of that technology.

The third way to quantify system performance is simulation, described in more detail in Sec. 5.4.2. Because it is time-consuming, we typically use simulations only for key performance parameters. However, simulations allow much more complex modeling and can incorporate limits on performance from multiple factors (e.g., resolution, stability, and data rate). Because they typically provide much less insight, however, we must review the results carefully to see if they apply to the specific situation we have in mind. Still, in complex circumstances, a simulation may be the only acceptable way to quantify system performance. A much less expensive method of simulation is the use of commercial mission analysis tools or even Excel or MATLAB models.

5.4 Mission Utility and Figures of Merit—Is the Mission Worthwhile? (Step 9)

Is the mission worth doing? Does it meet the fundamental objectives of the end user or the people who paid to create it? Does FireSat II save lives and property and reduce fire fighting costs sufficiently to warrant flying the mission? *Mission utility analysis* is the process of trying to provide a quantitative assessment of the degree to which the system meets the basic objectives it is attempting to fulfill. This is used to (1) provide quantitative information for decision making, and (2) provide feedback on the system design. Ultimately, an individual or group will decide whether to build a space system and which system to build based on an overall assessment of whether the system can meet its mission objectives at an acceptable cost and risk and do so better than alternative approaches. As discussed in Sec. 5.7, this does not mean the decision is, or should be, fundamentally technical in nature. However, even though basic decisions may be political, economic, or sociological, the best possible quantitative information from the mission utility analysis process should be available to support them.

Mission utility analysis also provides feedback for the system design by assessing how well alternative configurations meet the mission objectives. FireSat II shows how this process might work in practice. Mission analysis quantifies how well alternative systems can detect and monitor forest fires, thereby helping us to decide whether to proceed with a more detailed design of several satellites in low-Earth orbit or a single larger satellite in geosynchronous orbit. As we continue these trades, mission utility analysis establishes, for example, the probability of being able to detect a given forest fire within a given time, with and without FireSat II, and with varying numbers of spacecraft. For FireSat II, the decision makers are those responsible for protecting the forests of the United States. We want to provide them with the technical information they need to determine whether they should spend their limited resources on FireSat II or on some alternative. If they select FireSat II, we will provide the technical information needed to allow them to select how many satellites and what level of performance to provide.

5.4.1 Performance Parameters and Measures of Effectiveness

The purpose of mission analysis is to quantify the system's performance and its ability to meet the ultimate mission objectives. Typically this requires two distinct types of quantities—performance parameters and measures of effectiveness. *Performance parameters*, such as those shown in Table 5-6 for FireSat II, quantify how well the system works, without explicitly measuring how well it meets mission objectives. Performance parameters may include coverage statistics, power efficiency, or the resolution of a particular instrument as a function of nadir angle and altitude. In contrast, *measures of effec-*

Table 5-6. Representative Performance Parameters for FireSat II. By using various performance parameters, we get a better overall picture of our FireSat II design.

Performance Parameter	How Determined
Smallest detectable fire	Analysis
Resolution	Analysis
Orbit average area coverage rate	Simulation
Fire location accuracy	Analysis
Temperature of the fire	Analysis
Average number of times per day the fire is observed	Simulation

tiveness (MoEs) or *figures of merit* (FoMs) quantify directly how well the system meets the mission objectives. For FireSat II, the principal MoE will be a numerical estimate of how well the system can detect forest fires or the consequences of doing so. This could, for example, be the probability of detecting a given forest fire within 6 hours, or the estimated dollar value of savings resulting from early fire detection. Table 5-7 shows other examples for FireSat II.

Table 5-7. Representative Measures of Effectiveness (MoEs) for FireSat II. These Measures of Effectiveness help us determine how well various designs meet our mission objectives.

Goal	MoE	How Estimated
Fire Detection	Probability of detection vs. time since start of fire (milestones at 6, 12, and 24 hours)	Simulation
Prompt Knowledge	*Time late*—time from observation to availability to the fire captain	Analysis + Budget
Containment	Probability of containment vs. time	Simulation
Save Property and Reduce Cost	Value of property saved plus savings in fire fighting costs	Simulation + Analysis
Save Lives	Estimate of the number of lives saved as a result of having FireSat II available	Scenario Evaluation

We can usually determine performance parameters unambiguously. For example, resolution is determined by an optical analysis as discussed in Sec. 17.3 and the level of coverage for any point on the Earth's surface is determined by analysis or simulation. The probability of detecting and containing forest fires (i.e., MoEs) better measure our end objective, but are also much more difficult to quantify. It may depend on how we construct scenarios and simulations, what we assume about ground resources, and how we use the FireSat II data in conjunction with other data sources to fight fires.

Clearly, performance parameters and MoEs are unique to each mission and depend on the particular objectives that we are trying to achieve and constraints that we must live with. Nonetheless, in general, we would like to minimize cost and risk, maximize the performance, and obtain the results in a timely manner. Building a satellite to help the warfighter in a specific war is of very little utility if the war is over by the time the satellite is launched. Table 5-8 lists some of the most common performance parameters that are likely to be useful in evaluating mission utility for any mission. Of course, many of these will not apply to a specific mission and we are likely to need some additional mission-specific parameters that aren't on the list, but the table provides a reasonable place to start.

Good measures of effectiveness are critical to successful mission analysis and design. If we cannot quantify the degree to which we have met the mission objectives, there is little hope that we can meet them in a cost-effective fashion. The rest of this section defines and characterizes good measures of effectiveness, and Sec. 5.4.2 shows how we evaluate them.

Good measures of effectiveness must be

- Clearly related to mission objectives
- Understandable by decision makers
- Quantifiable
- Sensitive to system design (if used as a design selection criterion)

MoEs are useless if decision makers cannot understand them. "Acceleration in the marginal rate of forest-fire detection within the latitudinal coverage regime of the end-of-life satellite constellation" will likely need substantial explanation to be effective. On the other hand, clear MoEs which are insensitive to the details of the system design, such as the largest coverage gap over one year, cannot distinguish the quality of one system from another. Ordinarily, no single MoE can be used to quantify how the overall system meets mission objectives. Thus, we typically want to provide several MoEs summarizing the system's capacity to achieve its broad objectives.

Measures of effectiveness generally fall into one of three broad categories associated with (1) discrete events, (2) coverage of a continuous activity, or (3) timeliness of the information or other indicators of quality. Discrete events include forest fires, nuclear explosions, ships crossing a barrier, or cosmic ray events. In this case, the best measures of effectiveness are the rate that can be sustained (identify up to 20 forest fires per hour), or the probability of successful identification (90% probability that a forest fire will be detected within 6 hours after ignition). The probability of detecting discrete events is the most common measure of effectiveness. It is useful both in providing good insight to the user community and in allowing the user to create additional measures of effectiveness, such as the probability of extinguishing a forest fire in a given time.

Some mission objectives are not directly quantifiable in probabilistic terms. For example, we may want continuous coverage of a particular event or activity, such as continuous surveillance of the crab nebula for extraneous X-ray bursts or continuous monitoring of Yosemite for temperature variations. Here the typical measure of

Table 5-8. Performance Parameters Applicable for Most Space Missions. For additional details, see Wertz [2008]. These are primarily technical measures that can be evaluated analytically or with a mission simulation.

1. Cost

Primary Performance Parameters

Non-Recurring Engineering (NRE) = full cost of developing the system

Recurring Cost per Mission = ongoing cost of building and launching subsequent units, including launch, spacecraft bus, payload, and operations (often normalized to 1 year)

Secondary Performance Parameters

Cost per Year—not meaningful if the goal is to achieve a particular objective such as return of samples from a comet

Cost per Unit of Performance—here "performance" should be one of the goal-oriented MoEs. ***Cost per data bit*** is a bad idea in most cases—it's results we want, not data volume.

2. Coverage

Observations per Day—may be per satellite or for the system as a whole, if there are multiple satellites

Number of Spacecraft = number of spacecraft required to achieve a given level of coverage or frequency of observations. May change with time as more spacecraft are added to the system.

3. Data Quality or Quantity

Resolution at Nadir—See Sec. 17.3 for definitions and formulas. An alternative measure often used by image analysts is the ***National Imagery Interpretability Ratings Scale (NIIRS)*** that is based on the ability to interpret the resulting images (See Fiete [1999].)

Bandwidth or ***Number of Simultaneous Users***—as discussed throughout the book, there is an ever-increasing demand for more bandwidth

4. Mission Risk

Probability of Successful Deployment—takes into account both launch failures and failure to work on orbit when first deployed

Probability of a Successful Mission—will be much higher if a back-up is available in case of a launch or on-orbit failure

Ability to Reconstitute Assets—note that the back-up system may well be of lesser capability than the primary one which it is replacing. Most important in military missions or human resupply.

5. Flexibility

Orbit Agility—having a high delta V available provides much higher utility for many missions because of the capacity to adjust the orbit to meet changing mission needs

Observation Time Flexibility —this allows the "driver" to adjust both conditions and arrival time at any given location

Multi-Mission Utility = potential to use one asset for more than one mission

6. Responsiveness

Total Response Time (TRT) = total time from when a new request for data is made until the data is delivered to the end user, including launch and on-orbit check out

Mean Response Time = mean time from when a new request for data is made to an on-orbit asset until the data is delivered to the end user

Time Late = time from when an observation is made until the data is delivered to the end user. Measures how "stale" the delivered data is

Development Time = total time to develop, build, and launch a new satellite

effectiveness is some type of coverage or gap statistics such as the mean observation gap or maximum gap under a particular condition. Unfortunately, Gaussian (normal probability) statistics do not ordinarily apply to satellite coverage; therefore, the usual measure of average values can be very misleading. Additional details and a way to resolve this problem are part of the discussion of coverage measures of effectiveness in Sec. 10.2

A third type of measure of effectiveness assesses the quality of a result rather than whether or when it occurs. It may include, for example, the system's ability to resolve the temperature of forest fires. Another common measure of quality is the timeliness of the data, usually expressed as time late, or, in more positive terms for the user, as the time margin from when the data arrives until it is needed. Timeline MoEs might include the average time from ignition of the forest fire to its initial detection or, viewed from the perspective of a potential application, the average warning time before a fire strikes a population center. This type of information, illustrated in Fig. 5-2 allows the decision maker to assess the value of FireSat II in meeting community needs.

The space mission engineer is unlikely to know from looking at Fig. 5-2 whether this FireSat II data will have a high level of utility to fire fighters. But it is likely that someone who manages fire fighting activities and distri-

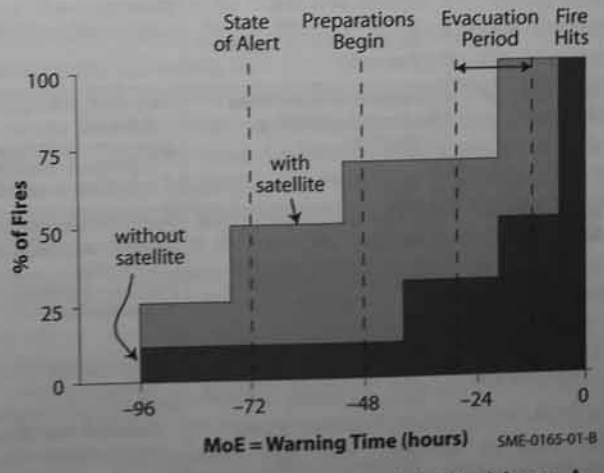


Fig. 5-2. Forest Fire Warning Time for Inhabited Areas. A hypothetical measure of effectiveness for FireSat II.

bution of fire fighting resources, will understand clearly the level of value that additional warning time brings. It may be that the level of advanced warning provided by FireSat II will allow equipment to be put in place and people evacuated in time to make a major difference in the cost and risk of fighting some fires, or it may be that the extra time has very little value if all that is needed is sufficient time for evacuation. Irrespective of the result, the type of information shown in Fig. 5-2 is what is needed to truly judge the utility of FireSat II and, therefore, whether it is worth the cost.

Because MoEs measure the ability of the system to satisfy the mission objectives, they are typically much more specific to each individual mission. In most cases, the objectives of an educational mission or an alt-space mission are very different than those of a military or a commercial mission, as discussed in Sec. 2.1. Most of the mission-specific MoEs fall into 4 broad categories reflecting the impact of the system on the ultimate mission objectives:

- ***Measures of Utility*** = how useful the mission is at meeting the objectives
- ***Measures of Success*** = how successful the mission is at meeting its objectives
- ***Impact on Outcome*** = whether the mission changes the ultimate outcome of events sufficiently to justify its cost
- ***Level of Preparedness*** = whether the mission allows us to be better prepared for future events

Table 5-9 gives some examples of these *goal-oriented MoEs* for several of the space mission communities identified in Chap. 2. As with performance parameters, the selection of specific MoEs depends on the mission objectives. In identifying MoEs, we want to look carefully at the mission objectives and ask what a good measure would be of the level at which the mission achieves these objectives.

5.4.2 Evaluating Mission Utility

In analyzing mission utility, we try to evaluate the measures of effectiveness numerically as a function of cost and risk, but this is hard to do. Instead, we typically use principal system parameters, such as the number of satellites, total on-orbit weight, or payload size, as stand-ins for cost. Thus, we might calculate measures of effectiveness as a function of constellation size, assuming that more satellites cost more money. If we can establish numerical values for meaningful measures of effectiveness as a function of the system drivers and understand the underlying reasons for the results, we will have taken a major step toward quantifying the space mission engineering process.

Recall that mission utility analysis has two distinct but equally important goals—to aid design and provide information for decision making. It helps us improve the mission design by examining the relative benefits of alternatives. For key parameters such as payload type or overall system power, we can show how utility depends on design choices, and therefore, intelligently select among design options.

Mission utility analysis also provides information that is readily usable by decision makers. Generally those who determine funding levels or whether to build a par-

Table 5-9. Goal-Oriented Measures of Effectiveness (GO-MoEs). Columns 2, 3, and 4 are representative space communities from Chap. 2. Column 5 is a specific example of a type of application mission.

Category	Military	Science	Education	Disaster Monitoring
Measures of Utility	How will the mission impact our ability to predict, protect, respond, retaliate, restore, rescue, contain, or limit collateral damage	Amount and importance of scientific data collected Ability to collect data not previously possible	Ability to design, build, test, and launch student experiments Cost of educational space experiments	Accuracy and speed of damage assessment Speed of getting the data to the people that need it
Measures of Success	Reduction in Casualties Reduction in combat duration	Detection of things never before detected Discovery of objects or processes that were previously unknown or not understood	Number of student experiments launched Number launched while the student is still a student	Rapidly locating and assessing damage Finding people in need of help or rescue in time to help or rescue them
Impact on Outcome	Change the outcome of the battle or war Tip the balance against future terrorist attacks	Prove or disprove theories Develop new scientific ideas or concepts Discover previously unknown phenomena	Quality of astronautics and space science education Number of students going into astronautics or space science	Number of lives saved Value of property saved Reduction in amount or duration of human suffering (e.g., getting food quicker to where it is needed)
Level of Preparedness	Ability to respond to changing world events or terrorist attacks Ability to prevent future conflicts by having advanced warning	Ability to respond to unexpected scientific events Ability to gain knowledge from transient events (e.g., nearby nova explosion)	Ability to educate the next generation of scientists and engineers Ability to demonstrate to children and the public the value of space science and astronautics	Ability to respond rapidly to the next major natural or man-made disaster Ability to prevent disasters (e.g., earthquake prediction; hurricane forecasting)

Table 5-9, Fig. 5-2◀, Eq. 5-0

ticular space system do not have either the time or inclination to assess detailed technical studies. For large space programs, decisions ultimately depend on a relatively small amount of information being assessed by individuals at a high level in industry or government. A strong utility analysis allows these high-level judgments to be more informed and based more on sound technical assessments. By providing summary performance data in a form the decision-making audience can understand, the mission utility analysis can make a major contribution to the decision-making process.

5.4.2.1 Mission Utility Simulation

Typically, the only effective way to numerically evaluate mission utility is to use a mission utility simulation designed specifically for this purpose. This is **not** the same as a *payload simulator*, which evaluates performance parameters for various payloads. For FireSat II, a payload simulator might compute the level of observable temperature changes or the number of acres that can be searched per orbit pass. In contrast, the *mission simulator* assumes a level of performance for the payload and assesses its ability to meet mission objectives. The FireSat II mission simulator would determine how soon forest fires can be detected or the amount of acreage that can be saved per year.

In principle, mission simulators are straightforward. In practice, they are expensive and time consuming to create and are rarely as successful as we would like. Attempts to achieve excessive fidelity tend to dramatically increase the cost and reduce the effectiveness of most mission simulators. The goal of mission simulation is to estimate measures of effectiveness as a function of key system parameters. We must restrict the simulator as much as possible to achieving this goal. Overly detailed simulations require more time and money to create and are much less useful, because long run times keep us from running them enough for effective trade studies. The simulator must be simple enough to allow making large number of runs, so we can collect statistical data and explore various scenarios and design options.

The mission simulation should include parameters that directly affect utility, such as the orbit geometry, motion or changes in the targets or background, system scheduling, and other key issues, as shown in Fig. 5-3. The problem of excessive detail is best solved by providing numerical models obtained from more detailed simulations of the payload or other system components. For example, we may compute FireSat II's capacity to detect a forest fire by modeling the detector sensitivity, atmospheric characteristics, range to the fire, and the background conditions in the observed area. A detailed payload simulation should include these parameters. After running the payload simulator many times, we can, for example, tabulate the probability of detecting a fire based on observation geometry, time of day, and size of the fire. The mission simulator uses this table to assess various scenarios and scheduling algorithms. Thus, the mission simulator might compute the mission geometry,

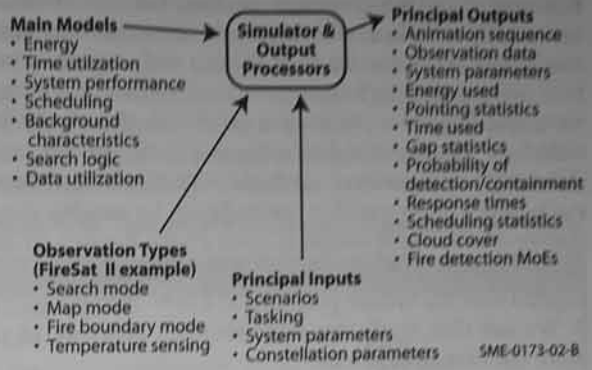


Fig. 5-3. Typical Mission Utility Simulation Inputs, Models, and Outputs for FireSat II.

time of day, and fire size, and then use the lookup table to determine the payload effectiveness.

With this method, we can dramatically reduce repetitive computations in each mission simulator run, do more simulations, and explore more mission options than with a more detailed simulation. The mission simulator should be a collection of the results of more detailed simulations along with unique mission parameters such as the relative geometry between the satellites in a constellation, variations in ground targets or background, and the system scheduling or downlink communications. Creating sub-models also makes it easier to generate utility simulations. We start with simple models for the individual components and develop more realistic tables as we create and run more detailed payload or component simulations.

In a constellation of satellites, scheduling is often a key issue in mission utility. The constellation's utility depends largely on the system's capacity to schedule resource use appropriately among the satellites. At the end of a single simulation run, the system should collect and compute the statistics for that scenario, generate appropriate output plots or data, and compute individual measures of effectiveness, such as the percent of forest fires detected in that particular run.

The next step is to run more simulations using new start times or otherwise varying the conditions for the scenarios. Changing the start times alters the relative timing and geometry between the satellites and the events they are observing, thus, averaging results caused by these characteristics. Collecting statistics on multiple runs is called a *Monte Carlo simulation*. For example, we might average the percentage of forest fires detected over different runs with different timing, but using the same scenario, to estimate the overall probability of detecting forest fires—our ultimate measure of effectiveness. The mission simulator should accumulate output statistics and prepare output plots over the Monte Carlo runs.

Frequently, in running mission simulations, we must choose between realistic and analytical scenarios. Realistic scenarios usually are too complex to help us understand how the system works but are still necessary to satisfy the end users. On the other hand, simple scenarios

illuminate how the system is working but do not show how it will work in a real situation. The best answer is to use simple scenarios for analysis and realistic scenarios to assess mission performance. In the FireSat II example, we might begin by studying a single satellite to determine how it behaves and then expand to a more complex simulation with several satellites. We might also start evaluating a multi-satellite constellation by looking at its response to a simple situation, such as one fire or a small group of uniformly distributed fires. This trial run will suggest how the system performs and how changes affect it. We can then apply this understanding as we develop more realistic simulations.

A related problem concerns using a baseline scenario to compare options and designs. Repeating a single scenario allows us to understand the scenario and the system's response to it. We can also establish quantitative differences by showing how different designs respond to the same scenario. But this approach tends to mask characteristics that might arise solely from a particular scenario. Thus, we must understand what happens as the baseline changes and watch for chance results developing from our choice of a particular baseline scenario.

Finally, mission simulations must generate usable and understandable information for decision makers—information that provides physical insight. Two examples are strip charts of various system characteristics and animated output. A *strip chart* plot is similar to the output of a seismograph or any multi-pin plotter, in which various characteristics are plotted as a function of time. These characteristics might include, for example, whether a particular satellite is in eclipse, how much time it spends in active observation, and the spacecraft attitude during a particular time step. Plots of this type give a good feel for the flow of events as the simulation proceeds.

A valuable alternative for understanding the flow of events is looking at an animation of the output, such as a picture of the Earth showing various changes in the target, background, and observation geometry as the satellites fly overhead. Thus, as Fig. 5-4 illustrates, an animated simulation of FireSat II output could be a map of a fire-sensitive region with areas changing color as fires begin, lines showing satellite coverage, and indications as to when fires are first detected or when mapping of a fire occurs. Animation is not as numerical as statistical data, but it shows more clearly how the satellite system is working and how well it will meet broad objectives. Thus, mission analysts and end users can assess the system's performance, strengths, and short-comings, and the changes needed to make it work better.

5.4.2.2 *Scenario Evaluation—The Hawaii Disaster of 2025*

Many of the MoEs in Table 5-9 are very difficult to evaluate numerically in large part because they aren't intended to be engineering oriented, but are meant to measure how well the mission objectives are being met. Realistically, there probably is no way to estimate via

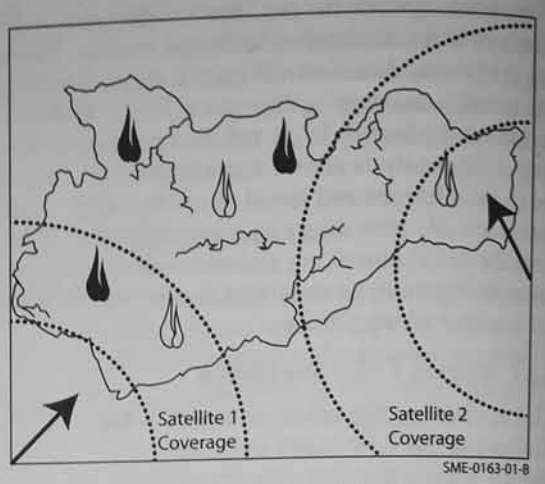


Fig. 5-4. Hypothetical Animation Output for FireSat II Mission Utility Simulator. Color displays are very valuable for animation sequences because we need to convey multiple parameters in each frame.

simulation the number of lives saved, the impact on the outcome of the war, or the importance of the resulting science. One of the best approaches to dealing with this is *scenario evaluation* in which we look at a real or hypothetical scenario and try to evaluate how that scenario would have been different had the system being evaluated been in place. Typically we would look to people who have been through the scenario if it's a real historical event, or similar ones if it's a hypothetical scenario, to estimate what the consequences would have been had the space system been in place. The advantage of using a real historical event or sequence of events is that at least for the baseline case, i.e., what happened without the system in place, the results are reasonably well understood. In addition, this type of evaluation is meaningful to decision makers because it is easy to discuss what actually happened and contrast that with what could have happened.

The scenario evaluation can be done either in conjunction with a simulation or, less precisely, by simply estimating the results. As an example, consider the great Hawaiian disaster of October 10, 2025 [Wertz, 2008]. On the evening of October 9, an earthquake and resulting tsunami struck in the ocean east of Japan with damage to all of the major airports and most homes, businesses, and infrastructure. Communications were badly disrupted and it was difficult to assess the damage or to find people that needed help, needed to be moved, had been washed out to sea, or were in imminent danger due to potential aftershocks. The tsunami also headed for Hawaii where evacuations of low ground began immediately. While major population centers were largely evacuated, it wasn't possible to get word to all of the residents and it was feared that many in rural areas may have been washed out to sea when the tsunami hit Hawaii.

By order of the President, HawaiiSat was launched from KSC 10 hours after the earthquake hit to provide both visible and IR images. The satellite carried an 0.5 meter diameter imager. Following a 4 deg inclination

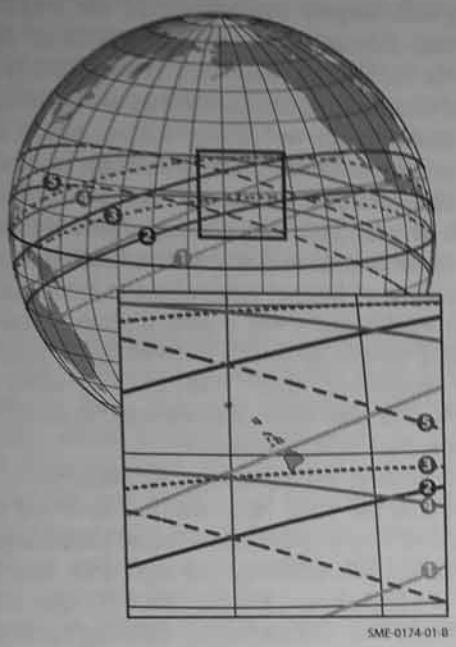


Fig. 5-5. Swath of the First 5 Orbits for the Hypothetical HawaiiSat Mission in October, 2025. Hawaii will be visible on 5 or 6 orbits/day with the observation times moving approximately 35 minutes earlier per day.

change at the equator, the satellite was placed in a 250 km altitude circular orbit at an inclination of 24.5 deg. The ground swath of HawaiiSat's first 5 orbits are shown in Fig. 5-5. The first images of the damage at a low elevation angle were returned on the first orbit about 80 minutes after launch. The second pass, 90 minutes later, flew almost directly over the islands and gave very good images of the damage and allowed the first estimate of the drift rate and direction of debris floating at sea by comparison with images from the first orbit. The next 2 orbits provided images at shallow angles for the big island and more nearly overhead for the rest of the islands. The fifth orbit again passed nearly overhead, 6 hours after the first orbit. 5 or 6 orbits of imaging occurred on subsequent days, moving about 35 minutes earlier per day.

The spacecraft parameter and cost estimates for the above system are based on the Frequent Revisit Surveillance mission discussed by Wertz et al. [2006] and Cox, et al. [2005, 2006]. Based on these parameters, the HawaiiSat performance parameters and MoEs are shown in Table 5-10.

While we don't yet have all of the numbers we need, it seems clear that a responsive disaster monitoring satellite could provide a significant capability to respond, restore, rescue, contain, and limit collateral damage. With this capability the government will be prepared to save lives and property.

In this type of scenario evaluation, some people will argue that existing on-orbit assets could do the job and that HawaiiSat wouldn't be needed. Others will argue that historically that doesn't happen and that existing assets are not in fact made available until well after the immediate need is past. Irrespective of which side of that argument prevails, the scenario evaluation approach, particularly when combined with real historical events and some mission simulations, can provide a great deal of insight into the utility of both existing and proposed systems and their strengths and weaknesses. This then leads to a more informed decision as discussed in Sec. 5.7.

Table 5-10. HawaiiSat Performance Parameters and MoEs.

Parameter	Value
Cost	
NRE	$40 M
Recurring Cost/Mission	$17.3 M
Coverage	
Observations/Day	5 to 6
Number of Spacecraft	1
Data Quality	
Nadir Resolution	0.3 m
Risk	
Deployment—Prob. of Success	90%
Mission—Prob. of Success	99%
Flexibility	TBD
Responsiveness	
Total Response Time	5 min
Mean Revisit Time	85 min for 6 to 7.5 hrs
Time Late	5 min
GO-MoEs	
Need a mission simulation and discussion with emergency response personnel to evaluate these critical measures	

5.5 Defining the Baseline Mission Concept, Revising Requirements, and Evaluating Alternatives (Steps 10–12)

Steps 10 to 12 in the Space Mission Engineering process from Table 3-1 are to define the baseline mission concept and architecture, revise the quantitative requirements and constraints, and evaluate alternatives. A *baseline mission*, like the point design defined in Sec. 5.2.3, is simply a mission definition that, in a broad sense, fulfills the overall mission objectives. By *mission definition* we mean what is being used for each of the 8 elements of the mission architecture plus the basic concept of operations. For FireSat II we want to know broadly how we're going to detect and monitor forest fires and how we're going to get that information to the end user. Ultimately, we are concerned with the firefighter in the truck and how he or she gets the data they need to do their job better and safer. Although we will work a great many technical details as we go through the definition of the mission, we should return often to the end user. Are we meeting their needs in a timely fashion at minimum cost and risk? If we aren't convinced that this is the case, then we should go back and work on the problem until we are. Space mission engineering is a highly iterative process and it is rarely the case that the

first approach is the best, and our first ideas may not even work that well.

Table 5-11 summarizes the process for selecting the baseline, trading on requirements, and evaluating alternative mission concepts. We begin by going back to the broad mission objectives and constraints that define what the mission is all about (Sec. 3.3). What is it that we're trying to achieve? Are there major constraints on how we are allowed to do that, such as the use of a particular technology or building on prior experimental missions? Who are the people that will ultimately pay for the mission, who are the people that will use the data that it generates, and what are their goals and objectives? This leads us to Step 2 where we identify the space community or communities that are being served. Irrespective of the details of a particular mission, a mission being done for the education community will almost certainly have a different set of objectives and constraints than will a military mission or a science mission meant to explore the depths of the Universe.

Table 5-11. Process for Selecting A Baseline Mission and Evaluating Alternatives. See text for discussion. See Sec. 5.6 for specific examples for the FireSat II and SCS mission.

Step	Where Discussed
1. List mission objectives and constraints	Sec. 3.3
2. Identify the space community being served and the end users and their broad objectives and approaches	Sec. 2.1
3. Determine which is the most important criteria for the mission—performance, cost, risk, or schedule	Sec. 1.1, Chap. 3
4. For each element of the mission architecture plus the mission concept, select an approach consistent with Step 3. This is a ***Point Design*** and the ***Initial Baseline***.	Sec. 4.1 to 4.3, 5.2.3
5. List major alternative approaches	Chap. 4
6. Look for the critical requirements that drive performance, cost, risk, or schedule (may be different for alternative approaches)	Sec. 4.4
7. For the baseline and each major alternative, begin the process of the Trading on Requirements	Sec. 5.5
8. Select a ***Revised Baseline*** and Major Alternatives for further analysis	—
9. Document and Iterate. This is a highly iterative process and documenting the reasons for choices is key.	—

The third step is to determine the most important criteria for the mission—performance, cost, risk, or schedule. Is the group that is funding the mission dramatically risk averse, such that we want to use largely processes and hardware that have worked in the past, or are they open to new ideas and approaches that might be able to dramatically drive down cost or provide a new level of performance? Throughout much of the history of the space program, the emphasis on most missions was avoiding risk, largely irrespective of the consequences on cost and schedule. (A common quote in the space industry is "If it works, all else is forgiven.") As discussed in Chap. 1, this has brought us to a space program that, in many cases, we can no longer afford and that doesn't generate results quickly enough to have sufficient utility to many end users. An asteroid warning system is of very little use if it is cancelled because of excessive cost or delayed until after an asteroid hits New York. Because of budget constraints in nearly all space programs worldwide, there is now a strong tendency to ask whether we can do things differently, or adopt hardware or processes from one of the other space communities or non-space applications, in order to significantly reduce cost or schedule.

Having created the framework for what we're trying to achieve and the basic rules within which we need to work, the next step is to define an initial baseline mission that can do the job, although perhaps with less performance or at a higher cost, greater risk, or longer schedule than we would like. We do this by defining each of the 8 elements of the mission architecture:

- Subject
- Payload
- Spacecraft Bus
- Ground Segment
- Mission Operations
- Orbit
- Command, Control, and Communications Architecture
- Launch Segment

Each of these is defined in more detail in Table 4-1. In addition to the elements of the mission architecture, we need to define the broad mission concept—i.e., how all of the mission elements work together to meet the needs of the end user. As discussed in Secs. 4.2 and 5.6, the mission concept for FireSat II revolves primarily around how (and where) we identify a forest fire and distinguish it from a really big barbecue and how we get that information to the firefighter in the truck. The method of doing these assessments are defined in Chap. 4 and in the discussion of the quick, limited detail studies (feasibility assessment, sizing estimate, and point design) in Sec. 5.2.

In addition to the initial baseline, we also want to list the major alternatives. These will nearly always involve one of elements of the mission architecture or the mission concept. For FireSat II one of the major alternatives will be the orbit. Do we use one or more small satellites in LEO or a single large satellite in GEO? We will get at the process of numerically evaluating these alternatives shortly, but in this step we simply want to identify them. This is a part of the creative process of space mission engineering. What are the alternative approaches that we might use to increase performance or drive down cost,

risk, or schedule? Are there processes or hardware from other space communities or from the non-space arena that might work well for this mission? Do the advances in computing power and image recognition and analysis software allow us to identify forest fires on-board the spacecraft and go directly from the spacecraft to the end user or the fire management station? We select as our initial baseline the approach that appears to be most consistent with what is important to this user—performance, cost, risk, or schedule.

In order to evaluate alternative approaches, we first identify the *critical requirements* that serve to drive the performance, cost, risk, and schedule. The process of identifying these critical requirements is defined in Sec. 4.4. Note that both the critical requirements and the system drivers may be different for different approaches. Two key issues here are to define the **reasons** for each requirement and to look for requirements that drive the cost, but don't improve performance, risk, or schedule. **Our goal is to adjust both the system parameters and the mission requirements to achieve our mission objectives at the minimum cost, risk, and schedule, or at least within a cost, risk, and schedule that we can afford, given that cost and schedule are likely to increase as the mission definition proceeds.**

The process of mission definition is much like an individual buying a new car. What we want to buy is a combination of what we would like and what we can reasonably afford and is likely to be strongly influenced by what is available in the marketplace. Is there new, lightweight mirror technology that can help us build a larger telescope at much lower mass and cost than was possible the last time we built one? Will new computer technology let us do more data processing on board such that we can simply send down the results and not all of the data needed to get there? What we want is to help the customer become an intelligent consumer.

The numerical processes for evaluating the major alternatives are the system trade studies, performance assessments, and mission utility analyses defined in detail in Secs. 5.3 and 5.4. The performance parameter trades are typically straightforward analyses that are relatively easy to do. These can help greatly when we are looking at a simple trade like orbit altitude or the amount of power or the resolution that is required to do the mission. Mission utility assessments are likely to be much harder, but also more useful. For FireSat II, when we compare the single geosynchronous satellite with multiple small satellites in LEO, the two alternatives will have very little in common, except the ability to detect forest fires. The system drivers and critical requirements will be different for both and they will have very different operational approaches and hardware. The only realistic way to make a comparison between them is to look at the mission utility that each can achieve as a function of cost, risk, and schedule.

5.6 Examples: FireSat II and SCS

As an example, we would like to go through the definition of a baseline system and the identification of major alternatives and system trades for both FireSat II and the SCS constellation defined in Sec. 3.2. As discussed in Sec. 3.3, the primary objective of FireSat II is **to detect, identify, and monitor forest fires throughout the United States, including Alaska and Hawaii in near-real time.** and for the Supplemental Communications System, SCS, it is **to provide low-cost, low-data-rate, real-time supplemental communications to existing military systems and aide in natural disaster relief with continuous connectivity to CONUS.** For both missions there are several possible secondary objectives, but these may change with the definition of the system and, therefore, are more important later when we begin the evaluation of alternatives. The principal constraints on both systems are those that are implied by the primary mission objectives—to be low cost and to be in place at the time that they are needed. In addition, we expect that military systems will want secure communications, while this is not likely to be of concern for the natural disaster relief mission.

For FireSat II, the principal community being served is the Earth applications community and the end user is the forest service and fire fighting community. Other than being regular users of GPS, they have no connection with space activity and no real interest in space per se. Thus, they will be interested in FireSat II only to the extent that it solves a problem that they can't otherwise solve (such as early detection of fires in remote areas) or solves problems more economically or substantially better than the traditional non-space solutions. If monitoring forest fires from space can get them data sufficiently quicker or better than other approaches such that they can more quickly contain fires, reduce fire fighting costs, and save property and lives (including reducing the risk to the firefighters themselves), then there will be some interest. But it is also likely that they will be wary of a system that costs 10's of millions of dollars or more (possibly much more) and that is well outside their area of expertise. There is likely to be a level of (possibly unspoken) concern on their part that FireSat II is more of a gimmick to support the space industry than a real and useful tool for identifying and fighting fires. This, in turn, means that clearly demonstrating mission utility to a group that fundamentally doesn't know or care about space will be critical.

In contrast, SCS is intended to serve two communities, both of which already use space assets and recognize a very real need for better communications in remote locations. The military community uses space assets for a great many functions and, along with the intelligence community, is one of the largest users, builders, and developers of space assets. The disaster relief community (part of the Earth applications community) works on a much lower budget, but has made good use of the Surrey-developed Disaster Monitoring Constellation,

Table 5-11◀, Fig. 5-5◀, Eq. 5-0

which provided the first images published in *Aviation Week* of the aftermath of Hurricane Katrina. In both cases, the end user is likely an individual in a remote area with limited communications equipment (much like the firefighter in the truck) who needs communications with the outside world as reliably and continuously as possible to save lives and accomplish their basic mission.

For all 3 groups involved (fire fighting, disaster relief, and military), low-cost is the most important criteria for both FireSat II and SCS. The fire fighting and disaster relief communities have very little budget to devote to "external" systems, such as space missions. The military community has substantial resources for communications, but already has large, expensive systems in place. The only reason for SCS is to provide additional communications at low cost. Because the systems will be low-cost and most likely SmallSats, they do not need to be as risk averse as traditional, large, multi-billion dollar satellites. Nonetheless, risk is important because the budget is limited and replacing the systems would be painful.

Defining an initial baseline for each system will require looking at each of the 8 mission elements and the mission concept. Assuming we have done this initial analysis using each of the chapters defined in Sec. 4.1, the preliminary results for FireSat II and SCS are shown in Table 5-12. In some cases, the likely lowest cost approach has been selected as the baseline (the payload and mission concept for FireSat II and the orbit for SCS) and in some cases a more traditional approach has been selected as the baseline (the ground station approach for both). So long as we continue the system trade and system evaluation process, the choice of the initial baseline isn't critical. The baseline will evolve as we continue to make trades and evaluate alternatives. The risk here is that we may try to fix the baseline in concrete and treat it as somehow better than the alternative approaches and this is what we very much need to avoid. After a few rounds through the system trade process we will learn more about which approaches are better and what the real goals and objectives are, assuming that the customer or end user are a part of the trade process, as they should be.

Our next step is to identify the critical requirements that will tend to drive performance, cost, risk, and schedule. These will tend to be different for the different alternatives. For the baseline FireSat II mission, the critical requirements for performance, cost, risk, and schedule are most likely the revisit interval since that drives the number of satellites, and the mission concept, since that determines the complexity of both ground and on-board processing. If we look at the GEO option for FireSat II, revisit interval won't really matter since we can see CONUS continuously. The requirements that will matter for GEO are the resolution (which could lead to needing a bigger telescope) and the need to cover

Table 5-12. Definition of the Initial Baseline and Major Alternatives for FireSat II and SCS. While alternatives exist for essentially all of the elements, we are only concerned here with the most important alternatives that will have a major impact on performance, cost, risk, schedule, or the mission concept.

Element	FireSat II Baseline	FireSat II Major Alternatives	SCS Baseline	SCS Major Alternatives
Subject	Heat from fire	—	Handheld receiver	Cell phone with wire antenna to boost signal
Payload	IR sensor	Possibly multiple spectral bands	Software defined radio; bands TBD	—
Spacecraft Bus	SmallSat bus, body-mounted arrays, integral propulsion, Earth pointing	May want steerable payload to look specifically at forests	200 kg SmallSat bus with deployable arrays and fixed crosslinks fore and aft	Mass of the spacecraft can be a continuous variable depending on the equipment and launch
Ground Segment	Dedicated ground station in CONUS	Commercial, pay-per-pass ground system	2 dedicated ground stations for redundancy	Commercial, pay-per-pass ground system
Mission Operations	Single facility doing both fire detection and commanding	Single facility doing commanding and monitoring of the on-board fire detection	Single facility doing both payload and constellation control	—
Command, Control, and Communications Architecture	Direct link from ground segment to FireSat II	—	Continuous link from ground station in CONUS to all of the satellites	—
Orbit	2 satellites in LEO orbit at inclination = 60 deg	1 satellite in GEO stationed over CONUS	3 satellites in circular, equatorial orbits at 20,000 km; supplemented with 3 more equatorial or 3 polar	TBD satellites in elliptical orbits—not as good, but more traditional
Launch Segment	Pegasus or Minotaur I	Secondary launch on a larger vehicle	All 3 satellites launched on a single EELV	Single launches on Pegasus or Minotaur 4
Mission Concept	Fire detection by trained analysts at mission ops facility; fire data relayed back through FireSat II and down to end user	On-board fire detection and monitoring sent directly to end user with data going from time-to-time to monitoring activity	Fixed crosslinks provide continuous connectivity to CONUS with only 1 hop from anywhere within 50 deg of the equator	—

northern Alaska, which may be too far north to be easily visible from GEO. If GEO becomes a viable option as the analysis proceeds, we may choose to argue that forest fires are less of a problem in northern Alaska and, therefore, the latitude that we need to reach can be limited.

For the SCS baseline constellation the critical requirements will be the capability of the hand held receiver, the spacecraft bus mass (which determines the size and power of the payload), and the need to overcome jamming for the military mission since this could dramatically reduce the available bandwidth. In the SCS alternatives we have chosen to list the possibility of doing something to boost the gain of an ordinary cell phone so that either military personnel or relief workers could use their cell phone which they are much more likely to have with them than some extraneous piece of equipment, particularly in the case of emergency relief operations. It would be an enormous asset if we could do this, but we don't know if it can be done. Therefore, we will need to devote analysis time, and most likely quite a bit of test time, to see what is possible and reasonable to do under adverse conditions in remote areas.

For both FireSat II and SCS we have listed commercial, pay-per-pass ground stations as a potential low-cost option to significantly drive down the ground system and operations cost. In this model, data can be brought down and commands sent back up via any of a series of commercial ground stations around the world, typically for a few hundred dollars per pass. Data is then put on a secure internet link and, if needed, encrypted on board the spacecraft before being transmitted. The satellite system then becomes simply another node on the secure internet. This has the potential of being both lower cost and more reliable than dedicated ground stations, but it tends to be a non-traditional approach (although service provided ground stations have been in operation for decades) and, therefore, may be **perceived** as too high a risk.

As part of evaluating alternatives we will make use of all of the various analytic techniques defined in Secs. 5.2, 5.3, and 5.4 including performance assessments, systems trades, and utility analysis. For FireSat II we can do system trade studies for many of the alternative approaches:

- Use of single vs. multiple spectral bands
- Earth pointing vs. steerable payload
- Choice of ground station approach
- Launch vehicle options (we want to keep more than 1 option available in order to allow negotiation for the lowest price)
- Basic mission concept of ground vs. on-board identification and monitoring of forest fires

The key need of the mission utility analysis will be twofold—to evaluate the results of multiple satellites in LEO vs. a single satellite in GEO and to convince ourselves, the customer, and the end user that FireSat II will really have substantial mission utility and be worth the estimated cost and risk. This will, of course, depend very much on what the cost turns out to be.

For SCS we can probably do a simple performance assessment for the issue of spacecraft mass and system trades for the

- Orbit selection
- Choice of ground station approach
- Launch vehicle options (as with FireSat II, we want to keep more than 1 option available in order to allow negotiation for the lowest price)

The cell phone option will probably require substantial analysis and testing to see whether it is workable and may go forward or may be discarded as impractical. Here the main purpose of a utility analysis is to clearly show the customer (and possibly Congress) that the end result is well worth the probable cost.

Having done our systems trades and utility assessment we then go back and either revise or confirm our initial baseline approach. Here we need to do two things. We need to document the choices we have made and the studies and analysis that have led to them so that they can be revisited from time to time as needed. Second, and perhaps equally important, is to not believe our results. There will be many more trades and analyses as we go forward. We haven't even started yet to take into account potential secondary objectives for either mission and these may take on a more important role, particularly if one or more of the secondary missions becomes involved in funding the program. For example, GPS did not have sufficient political support to be funded when it first came about. For this reason the Nuclear Detection System (NDS) was added and provided the margin needed to generate sufficient support in Congress for the system to come about. This means that we must continue to remain open to trades and alternatives as the system becomes better defined. And we must always go back to the needs of the end user—the firefighter in the truck, the soldier in an isolated area, or the relief worker in a remote disaster area somewhere in the world. Are we truly satisfying their needs at minimum cost and risk?

5.7 Deciding Whether a Mission Should Proceed

This section is concerned not with the detailed engineering decisions for a space mission, but with the broad trades involved in defining the overall mission—whether to proceed with it and what concept to use. Decisions for space missions fall into three broad categories: (1) go/no-go decision on proceeding with the mission; (2) selection of the mission concept; and (3) detailed engineering decisions, which are generally described throughout this book.

In principle, the go/no-go decision depends on only a few factors, the most important of which are:

- Does the proposed system meet the overall mission objectives?

- Is it technically feasible?
- Is the level of risk acceptable?
- Are the schedule and budget within the established constraints?
- Do preliminary results show this option to be better than nonspace solutions?

In addition to the above technical issues, a number of nontechnical criteria are ordinarily equally or more important in the decision-making process:

- Does the mission meet its political objectives?
- Are the organizational responsibilities acceptable to all of the organizations involved in the decision?
- Does the mission support the infrastructure in place or contemplated?

For example, a mission may be approved to keep an organization in business, or it may be delayed or suspended if it requires creating an infrastructure perceived as not needed in the long term. The mission analysis activity must include nontechnical factors associated with space missions and see that they are appropriately addressed.

The top-level trades in concept selection are usually not fully quantitative, and we should not force them to be. The purpose of the trade studies and utility analysis is to make the decisions as informed as possible. We wish to add quantitative information to the decisions, not quantify the decision making. In other words, we should not undervalue the decision-maker's judgment by attempting to replace it with a simplistic formula or rule.

Table 5-13 shows how we might try to quantify a decision. Assume that a system costs $500 million, but an improvement could save up to $300 million. To save this money, we could use option A, B, or C. Option A would cost $50 million, but the probability of success is only 70%; B would cost $120 million with 99% probability of success; C would cost $200 million with a 99.9% probability of success.

Which option should we select? The table gives the cost if successful, the cost if the improvement fails, and the expected values of both the cost and net savings. By numbers alone, we would select option B with an expected savings of $177 million. However, reasonable and valid cases can be made for both A and C. In option A, we risk only $50 million, and, therefore, are minimizing the total cost if the improvement succeeds or if it fails. In fact, the $620 million cost of failure for option B may be too much for the system to bear, no matter the expected savings. Option C provides a net savings of "only" $100 million, but its success is virtually certain. Although savings for this option are less dramatic, it does provide major savings while minimizing risk. In this case, we may assume the cost to be a fixed $400 million, with failure being so unlikely that we can discount it. Option B, of course, balances cost and risk to maximize the expected savings.

Table 5-13. Mathematical Model of a Hypothetical Decision Process for a New Space System (costs in $M). Numerically, we would choose B or A' if it were available. As discussed in the text, any of the choices may be best depending on the decision criteria.

Current Cost	$500M
Potential Savings if Improvement is Successful	$300M

Option	Cost of Improvement	Probability of Success	Total Cost if Successful	Total Cost if Failed	Expected Total Cost	Expected Savings
A	$50M	70%	$250M	$550M	$340M	$160M
B	$120M	99%	$320M	$620M	$323M	$177M
C	$200M	99.90%	$400M	$700M	$400.3M	$99.7M
A'	$50M	80%	$250M	$550M	$310M	$190M

Suppose, however, that option A had an 80% probability of success as shown in A', rather than the original 70% probability. In this case, the expected savings of A' would increase to $190 million, and would make it the preferred approach in purely mathematical terms. However, most individuals or groups faced with decisions of this sort are unlikely to change from option B to A' based solely on the increase in estimated probability to 80%. Their decisions are more likely to depend on perceived risk or on minimizing losses. Using non-mathematical criteria does not make the decisions incorrect or invalid, nor does it make the numerical values unimportant. We need quantitative information to choose between options but we do not have to base our decisions exclusively on this information.

As a second example, we can apply the results of utility analysis to concept selection for FireSat II. In particular, the number of satellites strongly drives the cost of a constellation. If we select the low-Earth orbit approach for FireSat II, how many satellites should the operational constellation contain? More satellites means better coverage and, therefore, reduces the time from when a fire starts until it is first detected. Consequently, one of our key parameters is the *time late*, that is, the time from when a fire starts until the system detects its presence and transmits the information to the ground. Figure 5-6 plots the hypothetical time late vs. the number of satellites for FireSat II. The details of such plots will depend on the latitude under consideration, swath coverage, altitude, and various other parameters. However, the characteristic of increasing coverage with more satellites eventually reaches a point of diminishing returns. This will normally be true irrespective of the coverage assumptions.

If we assume an initial goal for time late of no more than 5 hours, we see from the plot that a system of 6 satellites can meet this goal. Alternatively, a 4-satellite system can achieve a time late of 6 hours. Is the smaller time late worth the increased number of satellites and the money to

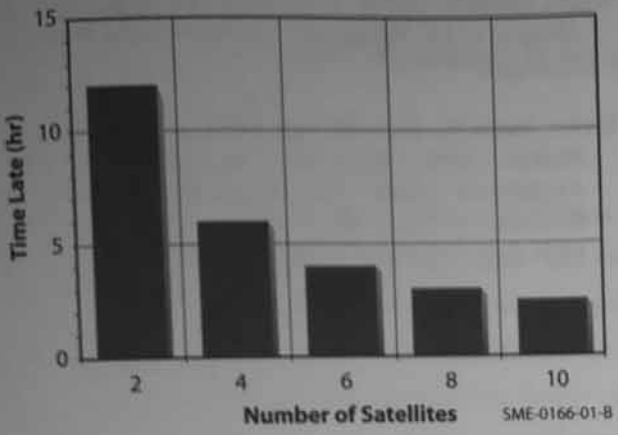


Fig. 5-6. Hypothetical Coverage Data for FireSat II. See text for definitions and discussion. As discussed in Sec. 10.10, growth in coverage typically comes in increments or plateaus. These are assumed to be 2-satellite increments for FireSat II.

build them? Only the ultimate users of the system can judge. The additional warning may be critical to fire containment and, therefore, a key to mission success. However, it is also possible that the original goal was somewhat arbitrary, and a time of **approximately** 5 hours is what is really needed. In this case, fire-fighting resources could probably be used better by flying a 4-satellite system with 6 hours time late and applying the savings to other purposes. Again, mission utility analysis simply provides quantitative data for intelligent decision making.

Of course, we must remember that the number of FireSat II satellites will depend not only on the utility analysis but also on politics, schedules, and resources. The public must see FireSat II as an appropriate response to the problem, as well as an efficient use of scarce economic resources compared to, for example, more fire fighters. In addition, a satellite system may serve several missions, with multiple mission criteria and needs. Just as we cannot apply only one criterion to some system drivers, we may not be able to balance numerically the several criteria for mission selection. Instead, the developers, operators, and users must balance them using the insight gained from the system trades and mission utility analysis.

Having undertaken a round of system trades, evaluated the mission utility, and selected one or more baseline approaches, we are ready to return to the issue of system requirements and their flow-down to various components. Chapter 6 treats this area, which is simply the next step in the iterative process of exploring concepts and defining requirements.

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